Holographic Coherence Relativity I: Coherence Frames, Frame Invariance, and the Born Rule

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We argue that the quantum measurement problem can be recast as a frame problem: whether a system appears coherent or definite depends on the coherence frame used to describe it, much as simultaneity depends on inertial frame in special relativity. We call the broader framework Holographic Coherence Relativity (HCR), while using "coherence relativity" for the frame-relativity principle developed in this paper; this avoids confusion with contextual realism and critical realism, whose abbreviations can overlap with bare "CR" in adjacent literatures. Physical reality is described on a joint manifold $\mathcal{M} \times \Sigma$, where $\mathcal{M}$ is emergent spacetime and Σ is the space of coherence frames. In this picture, wave function collapse is not a new dynamical process but a transformation between coherence frames induced by measurement interactions. We derive the Born rule for pure states as the unique probability measure invariant under coherence-frame relabelings and splittings, together with additivity and continuity, and outline the extension to mixed states by purification. This places insights from Gleason's theorem, Zurek's envariance, and operational reconstruction programs within a single geometric framework. We illustrate the proposal with the double-slit experiment in two coherence frames and identify eight experimentally testable signatures that distinguish HCR from standard quantum mechanics: non-linear visibility scaling, frame-transformation hysteresis, coherence-space curvature effects, modified gravitational decoherence, measurable transformation timescales, coherence revival, a Zeno/anti-Zeno crossover prediction, and the absence of preferred-basis selection without record constitution. The most accessible predictions—erasure hysteresis and transformation timescales—are testable with current superconducting-qubit and trapped-ion platforms.

I. INTRODUCTION

A. The Measurement Problem as a Frame Problem

The quantum measurement problem—how definite outcomes emerge from superposed states—has persisted for nearly a century despite the extraordinary empirical success of quantum mechanics. At its core lies a conceptual asymmetry: the formalism describes systems evolving unitarily, yet observations yield definite results governed by the Born rule, $p_k = |c_k|^2$. The standard resolution appends a measurement postulate that operates outside the unitary framework, creating what Penrose called "the most important open problem in physics" and what Bell termed "the shifty split" between quantum and classical domains [1].

Attempts to resolve this tension have proceeded along several lines. The mathematical framework for this tension was established by von Neumann [2]. Gleason's theorem [3] showed the Born rule follows from noncontextuality, but left its physical origin unexplained. Zurek's envariance [4, 5] derived probabilities from entanglement symmetry, though critics note potential circularity. The Deutsch–Wallace approach [6, 7] obtained Born weights from rationality axioms in an Everettian universe. The decoherence program [8, 9] explained interference suppression while assuming the Born rule. Relational quantum mechanics [10] and QBism [11] reinterpret the formalism without deriving probabilities. Objective collapse models [12–15] modify the dynamics itself.

The empirical decoherence literature also fixes the scale of the problem. Joos and Zeh showed that a dust grain in air decoheres on timescales of order 10^{-31} s, that even the best laboratory vacuum only stretches this to about 10^{-14} s, and that only near-intergalactic isolation pushes the timescale toward seconds; absent such monitoring, even macroscopic bodies such as Hyperion would sustain coherent orientation superpositions rather than stable classical trajectories [9, 16]. Haroche and collaborators later tracked this progressive decoherence directly in cavity QED [17, 18]. These facts matter here because they make classicality explicitly environment-relative, not an intrinsic property of an isolated system.

Each captures an essential aspect. Yet none simultaneously provides: (i) a derivation of the Born rule from physical principles, (ii) a geometric framework explaining *why* those principles hold, (iii) testable predictions, and (iv) ontological clarity about frame-invariant content.

B. Coherence as Relativity

We propose that the measurement problem is not a dynamical problem requiring new physics, but a *frame problem*—analogous to the shift from Newtonian absolute time to the relativity of simultaneity.

Throughout, we reserve HCR for Holographic Coherence Relativity. The shorter phrase "coherence relativ-

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ity" refers to the specific coherence-frame sector developed here, and we avoid the bare acronym "CR" because it is already overloaded in nearby philosophy-of-science literature.

The central idea: **coherence is frame-relative**. What appears as a definite outcome in one coherence frame appears as a coherent superposition in another, just as events simultaneous in one inertial frame are non-simultaneous in another. There is no absolute fact about whether a system "has collapsed."

We formalize this through four elements:

1. **Coherence space** Σ: A manifold whose points represent coherence frames. Physical reality is described on $\mathcal{M} \times \Sigma$. We treat $\mathcal{M}$ as a given Lorentzian manifold in this paper; a later paper in the series (Paper 3) will derive the spacetime manifold and its metric from the Hilbert-space coherence data; spectral geometry and quantum-chaos connections on Σ are developed in Paper 4.

2. **Frame transformations**: Maps $U_{F\to F'}$ between coherence frames, analogous to Lorentz transformations.

3. **Born rule as frame-invariant measure**: Requiring invariance under coherence relabelings and frame splittings, plus additivity and continuity, uniquely fixes $\mu(\psi_i) = |c_i|^2$.

4. **Tensor field** T_{AB}: A field on $\mathcal{M} \times \Sigma$ encoding the information geometry of coherence frames.

A coherence frame is more than a choice of orthonormal basis. A basis change is only a relabeling within a fixed descriptive setup. A coherence frame also specifies which degrees of freedom stably carry definite records under the relevant system–environment coupling. Formally, it is an equivalence class of bases together with a pointer observable selected by dynamical stability. The manifold Σ therefore labels physically different descriptive perspectives, not just different coordinate choices on the same description. Moving between frames has an information-geometric cost measured by the Fubini–Study metric and can, in general, depend on the path through Σ via the tensor field T_{AB}.

The main claim of this paper is structural rather than purely interpretive. We identify the coherence manifold $\Sigma = U(d)/T^d$ as the natural geometric arena for quantum description, and show that a frame-invariance principle on this space uniquely fixes the Born measure. On this view, collapse, coherence, and classicality are frame-dependent descriptions of one underlying reality, while probabilities are the quantities that remain unchanged across frames.

C. Organization

Section II introduces coherence frames and the geometry of Σ. Section III develops frame transformations and invariants. Section IV derives the Born rule. Section V resolves standard quantum paradoxes. Section VI discusses implications, and Sec. VII summarizes.

TABLE I. Principal notation. Symbols in order of first appearance.

Symbol	Meaning
Spaces	
$\mathcal{H}$	Hilbert space, dimension d
$\mathcal{M}$	Emergent spacetime (input in this paper; derived from $\mathcal{H}$ in
Σ	Coherence space $= U(d)/T^d$
Core objects	
$\mathcal{R}$	Physical reality (primitive)
$F \in \Sigma$	Coherence frame
$U_{F\to F'}$	Frame transformation (unitary)
$\mu_F(i)$	Probability measure on frame-F alternatives
Geometry	
G_{AB}, $G_{\lambda\lambda}$	Fubini–Study metric on Σ
Ω_{AB}	Frame-bundle curvature
T_{AB}	Coherence tensor $= G_{AB} + \Omega_{AB}$
λ	EGY distinguishability $\in [0,1]$
$s(\lambda)$	Accumulated geometric entropy (arc length)
$V(\lambda)$	Geometric quasipotential $= s(1) - s(\lambda)$
*Dynamics (*note disambiguations*)*	
F^A	Drift field on Σ *(distinct from frame F)*
$\mathcal{F}$	Free energy *(distinct from frame F)*
$\mathcal{S}[\gamma]$	Freidlin–Wentzell action
σ	Action path parameter *(distinct from arc s)*
V_{stoch}	Full stochastic quasipotential
$\mathcal{V}$	QM visibility $= \sqrt{1-\lambda^2}$ *(distinct from V)*
$\mathcal{K}_\Sigma$	Ricci scalar of Σ *(distinct from $\mathcal{R}$)*

II. COHERENCE FRAMES AND THE GEOMETRY OF QUANTUM DESCRIPTION

A. Definitions

Definition 1 (Coherence Space)**.** *Let $\mathcal{H}$ be a Hilbert space of dimension d. The* ***coherence space*** *Σ is the space of orthonormal bases of $\mathcal{H}$, modulo physically irrelevant phase redundancies:*

$$\Sigma = U(d)/T^d, \tag{1}$$

where $U(d)$ is the unitary group and T^d is the maximal torus. Mathematically, Σ is the complete flag manifold $\mathrm{Fl}(\mathbb{C}^d)$*, a compact Kähler manifold of real dimension d^2-d [19, 20]. The associated projective-state space $\mathbb{P}(\mathcal{H}) \cong \mathbb{CP}^{d-1}$ carries the standard Fubini–Study geometry [19–22]; in the present framework, the information geometry used on Σ is the pullback of that projective geometry along the state map determined by the chosen frame data.*

Throughout this paper, $\Sigma \equiv \Sigma_{\rm coh}(\mathcal{H}) = U(d)/T^d$ denotes the coherence manifold of $\mathcal{H}$. This is distinct from a string worldsheet $\Sigma_{\rm ws}$ and from configuration spaces such as $\mathrm{Map}(\Sigma_{\rm ws}, \mathcal{M})$; later papers will use the more explicit notation $\Sigma_{\rm coh}(\mathcal{H})$ whenever multiple meanings of Σ appear at once.

Definition 2 (Coherence Frame)**.** *A **coherence frame** $F \in \Sigma$ is a choice of preferred orthonormal basis $\{|i\rangle_F\}$ together with a specification of which degrees of freedom encode definiteness. Information never received by the system is excluded from its accessible sector on the same footing as information decohered to the bath.*

Remark 1 (Bloch sphere: state space versus frame space)**.** *For a qubit ($d = 2$), the coherence space $\Sigma = U(2)/T^2 \cong \mathbb{CP}^1 \cong S^2$ is geometrically the same manifold as the pure-state space $\mathbb{P}(\mathcal{H}) \cong \mathbb{CP}^1$. This coincidence can obscure an important conceptual distinction. As a state space, a point on the Bloch sphere specifies where the qubit state $|\psi\rangle$ is. As a coherence-frame space, a point specifies which orthonormal basis $\{|e_1\rangle, |e_2\rangle\}$ is taken to define definite alternatives. In coherence relativity, Σ always denotes the latter. A state $|\psi\rangle$ is represented in a frame $F = \{|e_1\rangle, |e_2\rangle\}$ by amplitudes c_1, c_2 with $|\psi\rangle = c_1 |e_1\rangle + c_2 |e_2\rangle$; the frame changes, while the frame-invariant Born weights remain $|c_i|^2$. For $d > 2$, the distinction is unavoidable because $\mathbb{P}(\mathcal{H}) = \mathbb{CP}^{d-1}$ and $\Sigma = U(d)/T^d$ are no longer the same manifold.*

Definition 3 (State Representation)**.** *For physical reality $\mathcal{R}$, the state in frame F is*

$$|\psi\rangle_F = \sum_i c_i^{(F)} |i\rangle_F , \tag{2}$$

where the coefficients $c_i^{(F)}$ depend on the frame. The same $\mathcal{R}$ has different representations in different frames: $|\psi\rangle_F(\mathcal{R}) \neq |\psi\rangle_{F'}(\mathcal{R})$ in general.

We take $\mathcal{R}$ as primitive: it denotes the physical situation being described, including the experimental arrangement, preparation, and relevant system–environment coupling. Operationally, $\mathcal{R}$ is the equivalence class of all frame-dependent descriptions $\{|\psi\rangle_F\}_{F\in\Sigma}$ related by frame transformations. The key extra ingredient is pointer structure: $\mathcal{R}$ includes which degrees of freedom stably encode definiteness. That is why $\mathcal{R}$ is not just a ray in Hilbert space. Two states related by a unitary that changes the pointer observable belong to different physical descriptions. In this sense, $\mathcal{R}$ lives on $\mathcal{M} \times \Sigma$ rather than in $\mathcal{H}$ alone, and its frame-invariant content is given by the quantities in Theorem 1: Born measures, inner products, spectra, and entanglement entropy.

Remark 2 (Frame manifold versus projective-state sector)**.** *Two related geometric spaces appear in the framework and should be kept distinct. The frame manifold $\Sigma = U(d)/T^d$ parameterizes ordered coherence frames, while the ray space $\mathbb{P}(\mathcal{H}) \cong \mathbb{CP}^{d-1}$ parameterizes pure states. Statements about isotropy, homogeneity, and the full Killing algebra apply directly to $\mathbb{P}(\mathcal{H})$ with its Fubini–Study metric; statements about preferred bases, pointer structure, and transport of definiteness apply to Σ. Coherence relativity links the two by treating Σ as the frame-bundle analog of the projective sector and by pulling the projective information geometry back to Σ through the state map. This distinction lets the theory preserve continuous unitary wave evolution on $\mathbb{P}(\mathcal{H})$ while using Σ to encode which descriptive frame renders stable records manifest.*

B. Examples

Single qubit. For $d = 2$, Σ is parameterized by the Bloch sphere. The frame F_z corresponds to $\{|0\rangle, |1\rangle\}$; the frame F_x to $\{|+\rangle, |-\rangle\}$. A state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in F_z becomes $|\psi\rangle = \frac{\alpha+\beta}{\sqrt{2}}|+\rangle + \frac{\alpha-\beta}{\sqrt{2}}|-\rangle$ in F_x. Coherence is not a property of the state alone but of the state-in-a-frame.

Double slit. The interference frame $F_{\rm int}$ has basis states that are path superpositions; the which-path frame $F_{\rm wp}$ has path eigenstates. Both describe the same setup (Sec. V).

Bipartite system. For $\mathcal{H}_A \otimes \mathcal{H}_B$, the global frame $F_{\rm global}$ exhibits entanglement while the local frame F_A shows only a mixed state. The reduced density matrix $\rho_A = \mathrm{Tr}_B[\rho]$ is the mathematical operation that extracts the F_A-description from the global one; the partial trace is not itself a frame transformation but rather the passage from one coherence-frame description to another. Decoherence—the progressive suppression of off-diagonal elements in a preferred basis—is reinterpreted geometrically as a trajectory through Σ from an entangled global frame toward a locally classical one.

Remark 3 (Composite systems, Killing flows, and uninterrupted wave evolution)**.** *For an N-partite system with total Hilbert space $\mathcal{H}_{\rm tot} = \bigotimes_{a=1}^N \mathcal{H}_a$ and $D = \prod_a d_a$, the relevant frame manifold is $\Sigma_{\rm tot} = U(D)/T^D$, while the corresponding pure-state sector is $\mathbb{P}(\mathcal{H}_{\rm tot}) \cong \mathbb{CP}^{D-1}$. The multiparticle Schrödinger equation*

$$i\hbar\,\partial_t |\Psi(t)\rangle = H |\Psi(t)\rangle \tag{3}$$

generates a one-parameter unitary family $U(t) = e^{-iHt/\hbar}$, whose induced action on the projective-state sector $\mathbb{P}(\mathcal{H}_{\rm tot})$ is an isometric, hence Killing, flow of the Fubini–Study geometry [21, 22]. Modulo overall phase, the corresponding Lie algebra is $\mathfrak{su}(D)$, so every admissible Hermitian Hamiltonian on the composite system defines a direction in the full algebra of possible unitary motions. When $H = \sum_a H_a$, the flow factorizes into subsystem motions; interaction terms make the flow non-factorizable, which is the geometric signature of entangling dynamics. In this sense, the projective sector for fixed Hilbert-space dimension carries the full class of ad-

missible unitary descriptions, while the actual physics selects the realized trajectory through it.

This preserves Schrödinger's "all-waveform, all-the-time" content at the level of the total state: the global wavefunction remains on a continuous unitary orbit of $\mathbb{P}(\mathcal{H}_{\rm tot})$ throughout. What changes in HCR is not the existence of the global wave, but which coherence frame renders stable records, quasi-classical sectors, and effective collapse manifest. At finite D, $\mathbb{P}(\mathcal{H}_{\rm tot})$ is homogeneous and isotropic under the Fubini–Study metric, while $\Sigma_{\rm tot}$ is the associated homogeneous space of ordered frames. If the emergent-spacetime program sketched for Paper 2 succeeds, one may then ask whether cosmological isotropy of $\mathcal{M}$ is inherited from this projective isotropy rather than introduced as an independent postulate. Whether an exact universal statement extends without qualification to the full universe, or to infinite-dimensional settings, is left to later work.

C. Metric Structure

The coherence space inherits the Fubini–Study metric:

$$G_{\lambda\lambda} = \mathrm{Re}[\langle\partial_\lambda\psi|\partial_\lambda\psi\rangle - \langle\partial_\lambda\psi|\psi\rangle\langle\psi|\partial_\lambda\psi\rangle]\,. \tag{4}$$

Equation (4) measures the information-geometric distance between nearby frames. The geodesic distance $d(F,F') = \inf_\gamma \int_\gamma \sqrt{G_{\lambda\lambda}}\,d\lambda$ gives the length of the shortest path in Σ connecting F to F'. The distinguishability parameter $\lambda \equiv \sqrt{1-|\langle W_L|W_R\rangle|^2}$ is the standard Englert–Greenberger–Yasin (EGY) coordinate [23, 24], so $\mathcal{V}^2+\lambda^2 = 1$ with $\mathcal{V} = |\langle W_L|W_R\rangle|$. Because the Fubini–Study metric measures statistical distinguishability [25], λ serves as a natural coordinate along the decoherence trajectory in Σ. *Convention note:* throughout the HCR series, λ always follows this EGY orientation: $\lambda = 0$ is the fully coherent (undistinguishable) endpoint, $\lambda = 1$ is the fully classical (pointer-state) endpoint. Paper 2 introduces a derived coupling parameter $\kappa(x,\xi)$—distinct from λ—that measures the strength of $\mathcal{M}$-Σ sector coupling; κ and λ are monotonically related but differ in orientation and domain.

Figure 1 shows $G_{\lambda\lambda}(\lambda)$ and its inverse $d\lambda/ds = 1/\sqrt{G_{\lambda\lambda}}$ for a multi-mode decoherence model ($N = 10$ collectively coupled environment modes). Under the EGY parameterization, the metric is finite at $\lambda = 0$ and diverges as $\lambda \to 1$. The universal value $G_{\lambda\lambda}(0) = 1/2$ means the fully coherent endpoint is geometrically accessible, whereas the divergence near $\lambda = 1$ expresses the rigidity of fully established classical records. For the single-mode analytic case ($N = 1$), $G_{\lambda\lambda} = 1/[2(1-\lambda^2)]$, and the multi-mode model has the same qualitative behavior. Accordingly, $d\lambda/ds = 1/\sqrt{G_{\lambda\lambda}}$ decreases from $\sqrt{2}$ at $\lambda = 0$ to 0 as $\lambda \to 1$: distinguishability changes quickly near the coherent endpoint and progressively more slowly as classical records stabilize. A coordinate-invariant and model-independent definition of the Markov transition is deferred to Paper 2 via the $\delta\lambda$ formalism.

D. The Tensor Field T_{AB}

The metric (4) captures the symmetric, information-geometric cost of moving between nearby frames. We now define the full tensor field T_{AB} on Σ that encodes both metric and connection structure.

Definition 4 (Coherence Tensor Field)**.** *Let x^A denote local coordinates on Σ, with $A, B, \ldots$ indexing coherence-space directions. The* ***coherence tensor field*** *is*

$$T_{AB}(x) = G_{AB}(x) + \Omega_{AB}(x), \tag{5}$$

where $G_{AB} = G_{BA}$ is symmetric and $\Omega_{AB} = -\Omega_{BA}$ is antisymmetric.

Symmetric part. The metric G_{AB} is the pullback of the Fubini–Study metric onto Σ along the state map $x \mapsto |\psi(x)\rangle$:

$$G_{AB}(x) = \mathrm{Re}[\langle\partial_A\psi|\partial_B\psi\rangle - \langle\partial_A\psi|\psi\rangle\langle\psi|\partial_B\psi\rangle]\,, \tag{6}$$

where $\partial_A := \partial/\partial x^A$. This is positive-semidefinite and measures the information-geometric cost of an infinitesimal frame change δx^A. Equation (4) is the one-dimensional restriction $G_{\lambda\lambda} = G_{AB}\,\delta^A_\lambda\,\delta^B_\lambda$ for the double-slit which-path parameter.

Antisymmetric part. The two-form Ω_{AB} derives from a connection A_A on coherence space, with curvature $F_{AB} = \partial_A A_B - \partial_B A_A$. It encodes non-integrable aspects of frame changes: how coherence frames twist relative to each other along closed loops in Σ. In quasi-equilibrium situations (reversible frame changes), $\Omega_{AB} = 0$ and T_{AB} reduces to the metric. For irreversible processes—such as partial erasure followed by attempted revival—$\Omega_{AB} \neq 0$ is the necessary condition for hysteresis: a closed loop in Σ with $\Omega_{AB} \neq 0$ fails to return the pointer observable to its initial configuration, so forward and reverse paths are geometrically inequivalent. The metric G_{AB} then determines the *magnitude* of the visibility deficit through the action cost (19); Ω_{AB} controls whether the deficit is nonzero at all.

The antisymmetric part Ω_{AB} has the same algebraic form as the Berry curvature two-form (Eq. (7)), but its geometric role here is distinct: it is the curvature of the *frame bundle* over Σ, encoding how the *pointer observable* (not the state vector) twists when coherence frames are transported along closed loops. Standard Berry curvature characterizes holonomy of the state vector under adiabatic parameter variation; Ω_{AB} characterizes holonomy of the pointer structure. Ω_{AB} vanishes for reversible (unitary) frame changes and is nonzero precisely when a closed loop in Σ fails to return the pointer observable to its initial configuration. The symmetric part G_{AB} reproduces the Fubini–Study metric—the Kähler metric on $\mathbb{CP}^{d-1}$—while Ω_{AB} encodes dynamical irreversibility in

Fig. 1 − Fubini−Study metric on Σ: accessible coherent endpoint, rigid classical endpoint

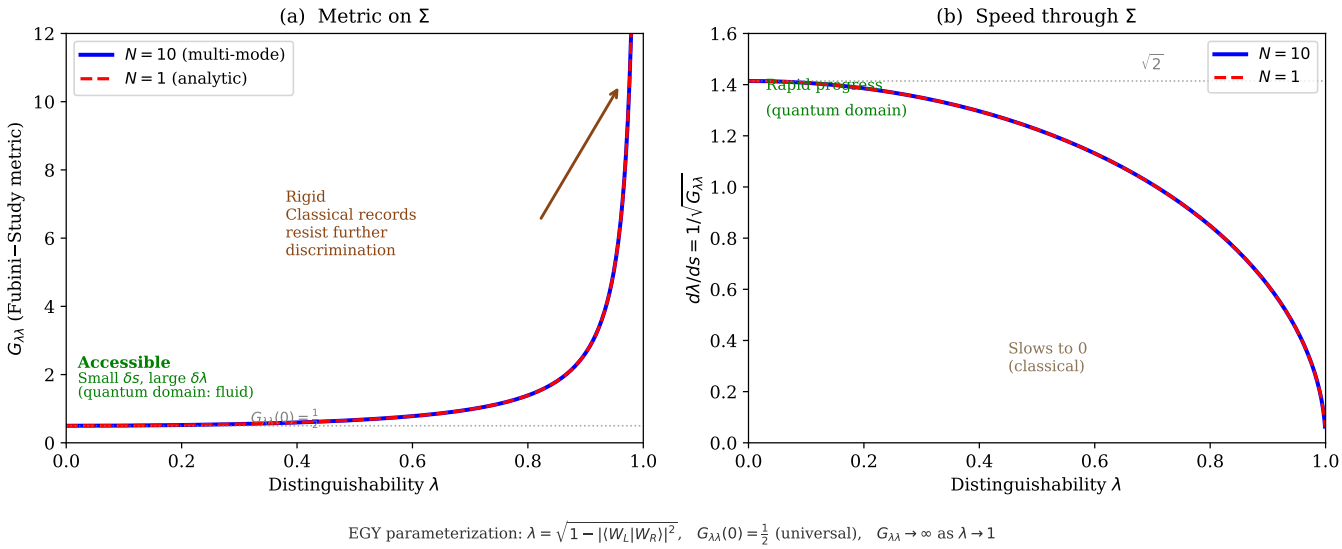

FIG. 1. Geometric structure of the decoherence trajectory under the EGY parameterization $\lambda = \sqrt{1 - |\langle W_L|W_R\rangle|^2}$ (multi-mode model, $N = 10$). **(a)** The Fubini–Study metric $G_{\lambda\lambda} = (ds/d\lambda)^2$: finite at $\lambda \to 0$ (all models: $G_{\lambda\lambda}(0) = 1/2$, geometric accessibility of the coherent frame) and diverging at $\lambda \to 1$ (rigidity of classical records). **(b)** The inverse $d\lambda/ds = 1/\sqrt{G_{\lambda\lambda}}$ starts at $\sqrt{2}$ and decreases monotonically: distinguishability accumulates efficiently near the coherent endpoint and slows as classical records solidify.

the frame bundle, a structure absent from the standard Kähler form. The full $T_{AB} = G_{AB} + \Omega_{AB}$ therefore carries genuinely new geometric content beyond standard projective Hilbert space structure. In Paper 2, Ω_{AB} is identified with the F_{ab} block of the imaginary part of $Q_{AB} = G_{AB} + iF_{AB}$; the reconciliation of the pointer-transport and state-vector holonomy perspectives is addressed there.

Concretely, the antisymmetric part takes the form

$$\Omega_{AB} = 2\,\mathrm{Im}\big[\langle\partial_A\psi|\partial_B\psi\rangle\big], \tag{7}$$

which is the Berry curvature two-form on projective Hilbert space when restricted to a single state; its role here as frame-bundle curvature is distinct because it encodes how the *pointer observable* (not the state vector) twists under transport on Σ.

Double-slit reduction. For the one-dimensional coherence manifold $\Sigma \simeq [0, 1]$ with which-path parameter λ, the tensor field reduces to the single scalar component $T_{\lambda\lambda} = G_{\lambda\lambda}(\lambda)$ (since $\Omega_{\lambda\lambda} = 0$ trivially in one dimension). The full tensor structure becomes physically relevant for systems with $\dim\Sigma \geq 2$, where closed loops in coherence space are possible.

The complete development of T_{AB}—including the decomposition into $\mathcal{M}$ and Σ coordinates, the cross-term $T_{M\Sigma}$ encoding $\mathcal{M}$-Σ coupling, the $\delta\lambda$ formalism separating frame-lag from decoherence progress, equations of motion, and the connection to emergent spacetime—is the subject of Paper 2. Here we require only the decomposition (5), the metric definition (6), and the qualitative role of Ω_{AB} in generating hysteresis. In Paper 2, T_{AB} is promoted to the quantum geometric tensor $Q_{AB} = G_{AB} + iF_{AB}$ on $\mathcal{M} \times \Sigma$; the Σ-Σ block of Q_{AB} recovers T_{AB}, and a new cross-term $T_{M\Sigma}$ (with no Paper-1 analog) encodes $\mathcal{M}$-Σ coupling. See Paper 2 for the precise correspondence.

E. Analogy with Special Relativity

The coherence-frame structure parallels the inertial-frame structure of special relativity (Table II). The key structural parallel: in SR, there is no absolute simultaneity—it depends on the inertial frame. In HCR, there is no absolute fact about whether a system is "in superposition" or "has collapsed"—it depends on the coherence frame.

III. FRAME TRANSFORMATIONS AND INVARIANTS

A. Frame Transformation Maps

Definition 5 (Frame Transformation)**.** *A **frame transformation** $U_{F\to F'}$ maps state representations: $|\psi\rangle_{F'} = U_{F\to F'}|\psi\rangle_F$. In the Hilbert space picture, $U_{F\to F'}$ is uni-*

tary:

$$|i\rangle_{F'} = \sum_j (U_{F\to F'})_{ji}\, |j\rangle_F \,. \tag{8}$$

Definition 6 (Coherence Relabeling)**.** *A **coherence relabeling** permutes and phase-rotates basis vectors:*

$$U_{F\to F'} : |i\rangle_F \to e^{i\theta_i}\, |\pi(i)\rangle_{F'} \,, \tag{9}$$

where π is a permutation and $\theta_i \in \mathbb{R}$.

B. Group Structure

Frame transformations form a **groupoid** over Σ:

1. Composition: $U_{F\to F''} = U_{F'\to F''} \circ U_{F\to F'}$
2. Identity: $U_{F\to F} = \mathbb{I}$
3. Inverse: $(U_{F\to F'})^{-1} = U_{F'\to F}$

For finite-dimensional $\mathcal{H}$, each $U_{F\to F'}$ has a unitary matrix representation preserving inner products: $\langle\psi|\phi\rangle_F = \langle U\psi|U\phi\rangle_{F'}$.

C. Transport, Alignment, and Fact Portability

The groupoid structure of frame transformations is not merely algebraic. It induces a notion of transport between frame-dependent descriptions of observables, records, and probabilities. This provides the kinematic basis for discussing when a fact remains stable across frames and when it does not.

For each coherence frame $F \in \Sigma$, we associate the data

$$F \equiv (\mathcal{O}_F,\, \mathcal{D}_F,\, P_F), \tag{10}$$

where $\mathcal{O}_F$ is the event algebra resolved in frame F, $\mathcal{D}_F \subset \mathcal{O}_F$ is the subset of events encoded as definite records in that frame, and P_F is a probability assignment on $\mathcal{O}_F$. At this stage P_F is left abstract; in Sec. IV we show that frame invariance uniquely fixes P_F to be the Born measure $\mu_F(i) = |c_i|^2$, so the compliance notion introduced below becomes Born compliance.

TABLE II. Structural analogy between special relativity and coherence relativity.

Special Relativity	Coherence Relativity
Spacetime event	Physical reality $\mathcal{R}$
Inertial frame	Coherence frame F
Coordinates $(t, \mathbf{x})$	State representation $\|\psi\rangle_F$
Lorentz transformation	Frame transformation $U_{F\to F'}$
Invariant interval ds^2	Born measure $\mu = \|c_i\|^2$
Minkowski metric	Fubini–Study metric on Σ
Simultaneity (relative)	Coherence (relative)
Temporal ordering (relative)	Past/present/future (relative)

Definition 7 (Transport Map)**.** *Given an allowed frame transformation $U_{F\to F'}$, the induced **transport map***

$$\mathcal{T}_{F\to F'} : \mathcal{O}_F \to \mathcal{O}_{F'} \tag{11}$$

is defined by conjugation:

$$\mathcal{T}_{F\to F'}(X) = U_{F\to F'}\, X\, U^\dagger_{F\to F'}, \qquad X \in \mathcal{O}_F. \tag{12}$$

In the present paper we restrict attention to transport induced by unitary frame transformations $U_{F\to F'}$. More general non-unitary transitions associated with coarse-graining, partial trace, or open-system evolution require an enlarged transport formalism and are deferred to Paper 2.

Definition 8 (Frame Alignment)**.** *Let $\mathcal{C} \subset \mathcal{O}_F$. Frames F and F' are **aligned on** $\mathcal{C}$ if:*

1. $\mathcal{T}_{F\to F'}(\mathcal{C}) \subset \mathcal{O}_{F'}$,
2. $\mathcal{T}_{F\to F'}(\mathcal{C} \cap \mathcal{D}_F) \subset \mathcal{D}_{F'}$,
3. $P_F(X) = P_{F'}(\mathcal{T}_{F\to F'} X)$ *for all* $X \in \mathcal{C}$.

Alignment means both event structure and probability assignment are preserved on $\mathcal{C}$.

Definition 9 (Measure-Compliant and Measure-Noncompliant Facts)**.** *A **frame fact** is an element $r \in \mathcal{D}_F$. The fact r is **measure-compliant** with respect to frame F' if: (i) r lies in a domain on which F and F' are aligned; (ii) $\mathcal{T}_{F\to F'}(r) \in \mathcal{D}_{F'}$; and (iii) $P_F(r) = P_{F'}(\mathcal{T}_{F\to F'} r)$. A fact is **measure-noncompliant** if any of these conditions fails.*

Remark 4. *After Theorem 2 identifies P_F with the Born measure $\mu_F(i) = |c_i|^2$, measure-compliant facts become **Born-compliant facts** and measure-noncompliant facts become **Born-noncompliant facts**. A Born-compliant fact is one that survives frame transport without loss of definiteness or change in invariant probability assignment.*

Definition 10 (Path Transport and Holonomy)**.** *Let $\gamma : F_0 \to F_1 \to \cdots \to F_n$ be a composable chain of frame transformations. The **transport along** γ is the ordered composition*

$$\mathcal{T}_\gamma = \mathcal{T}_{F_{n-1}\to F_n} \circ \cdots \circ \mathcal{T}_{F_0\to F_1}. \tag{13}$$

*Each $\mathcal{T}_{F_k\to F_{k+1}}$ acts on records $r \in \mathcal{D}_{F_k} \subset \mathcal{O}_{F_k}$ by the conjugation rule of Def. 7. If γ is a closed loop based at F_0, the associated map $H_\gamma := \mathcal{T}_\gamma$ is the **holonomy** of the loop.*

Proposition 1 (Observer-Independent Fact Criterion)**.** *Let $\mathcal{U} \subset \Sigma$ be an admissible transport domain and $r \in \mathcal{D}_F$ for some $F \in \mathcal{U}$. Then r defines an **observer-independent fact** on $\mathcal{U}$ if and only if: (i) r is measure-compliant with respect to every frame in $\mathcal{U}$; and (ii) for every closed loop γ in $\mathcal{U}$ based at F,*

$$H_\gamma(r) = r. \tag{14}$$

If there exists a loop for which $H_\gamma(r) \neq r$, then r is ***frame-relative****: locally definite in F, but not globally portable across $\mathcal{U}$.*

Remark 5 (Local and Universal Frames)**.** *Not all coherence frames are equally trustworthy across observers: a frame deep in the coherent regime carries records that few other frames can verify, while a fully decohered frame carries records that all frames agree on. The following four coordinates make this intuitive spectrum precise. The criterion above suggests a* universality spectrum *for coherence frames. Informally, the key questions are whether Born weights have already become definite records at all, and how far those records can be carried across other frames without changing their status. Four quantities are especially diagnostic: the decoherence degree λ, the richness of the record set $\mathcal{D}_F$, whether the Born measure is merely latent or instantiated in definite records, and the size of the admissible alignment domain on which those records remain measure-compliant and holonomy-trivial. Frames near the coherent limit ($\lambda \approx 0$) typically have $\mathcal{D}_F \approx \emptyset$ and only narrow alignment domains, so they are* local*: the Born measure is already defined, but not yet instantiated as portable fact. Frames near the pointer limit ($\lambda \to 1$) carry stable records and admit broad alignment domains on which relevant records satisfy Eq.* (14)*, so they are* universal *in the restricted sense of being globally portable across the decohered sector. The invariants themselves—Born measures, inner products, spectra, and entanglement entropies—do not lie on this spectrum; they are frame-invariant structure common to all frames. In this sense, the classical limit is the limit of maximal frame universality.*

Remark 6 (Relation to Bundle Curvature)**.** *The antisymmetric part Ω_{AB} of the coherence tensor field (Eq.* (7)*) has direct meaning in the transport language: nonzero curvature is the infinitesimal obstruction to path-independent transport of frame facts. On a region where $\Omega_{AB} = 0$, transport is flat and fact portability is path-independent. Where $\Omega_{AB} \neq 0$, nontrivial holonomy becomes possible, and frame-local definiteness need not extend to observer-independent facthood. Paper 2 develops this structure on the joint manifold $\mathcal{M} \times \Sigma$, where the cross-term $T_{M\Sigma}$ couples frame transport to spacetime dynamics.*

Remark 7 (Scope)**.** *The construction above establishes the kinematic notion of transport and the criterion for fact portability. Whether the resulting structure constitutes a true gauge symmetry—in which coherence-related descriptions are physically indistinguishable on all observables—depends on a redundancy test deferred to Paper 2. What is established here is a* relativity structure*: frame-local facts are real, Born-compliant facts are cross-frame portable, and Proposition 1 gives the precise criterion distinguishing the two.*

The paradox resolutions in Sec. V now have a precise form: apparent contradictions arise when a fact definite in one frame is transported outside its compliance domain, or around a loop with nontrivial holonomy. In that case there is no inconsistency in the theory; the contradiction lies in treating a frame-local fact as though it were a globally portable one.

D. Frame Splitting

Definition 11 (Frame Splitting)**.** *A* ***frame splitting*** *transforms a single basis state into an equal-amplitude superposition:*

$$|i\rangle_F \to \frac{1}{\sqrt{m_i}} \sum_{k=1}^{m_i} |i,k\rangle_{F'} \,, \tag{15}$$

where $\{|i,k\rangle_{F'}\}$ are orthonormal in the enlarged space.

Proposition 2 (Existence of Splittings)**.** *For any state $|\psi\rangle_F = \sum_i c_i |i\rangle_F$ with rational $|c_i|^2 = m_i/M$ (where $M = \sum_i m_i$), there exists a splitting $F \to F'$ such that $|\psi\rangle_{F'} = M^{-1/2} \sum_{i,k} |i,k\rangle_{F'}$ is equal-amplitude.*

Proof. Introduce an ancilla Hilbert space $\mathcal{H}_{\text{anc}}$ of dimension M with orthonormal basis $\{|e_j\rangle\}_{j=1}^M$. Partition the ancilla basis into blocks: for each i, assign m_i ancilla states $|e_j\rangle$ with $j \in B_i$, where $|B_i| = m_i$ and the blocks are disjoint with $\bigcup_i B_i = \{1, \ldots, M\}$.

Define an isometry $V : \mathcal{H} \to \mathcal{H} \otimes \mathcal{H}_{\text{anc}}$ by

$$V |i\rangle_F = \frac{1}{\sqrt{m_i}} \sum_{j \in B_i} |i\rangle_F \otimes |e_j\rangle \,. \tag{16}$$

Then V preserves inner products: $\langle i| V^\dagger V |i'\rangle = \delta_{ii'}$. Applying V to $|\psi\rangle_F$:

$$V |\psi\rangle_F = \sum_i c_i \cdot \frac{1}{\sqrt{m_i}} \sum_{j \in B_i} |i, e_j\rangle = \frac{1}{\sqrt{M}} \sum_i \sum_{j \in B_i} |i, e_j\rangle \,, \tag{17}$$

where the last equality uses $c_i = \sqrt{m_i/M}$ (up to phases absorbed by relabeling). Identifying $|i,k\rangle_{F'} \equiv |i, e_j\rangle$ for $j \in B_i$ gives the equal-amplitude state $|\psi\rangle_{F'} = M^{-1/2} \sum_{i,k} |i,k\rangle_{F'}$ in the enlarged frame F'. □

Physically, this construction corresponds to refining a measurement by coupling to an ancilla—a standard procedure in quantum information.

The frame splitting embeds Σ into the larger coherence space $\Sigma' = U(dM)/T^{dM}$; Axiom (A1) requires only that the probability measure μ is preserved under this embedding. We do not claim T_{AB} is defined on Σ'; the splitting is used solely to establish the probability assignment via Steps 2–3 of the Born rule derivation, after which the isometry returns to the original Σ. Compatibility of T_{AB} across dimension-changing embeddings is addressed in Paper 2.

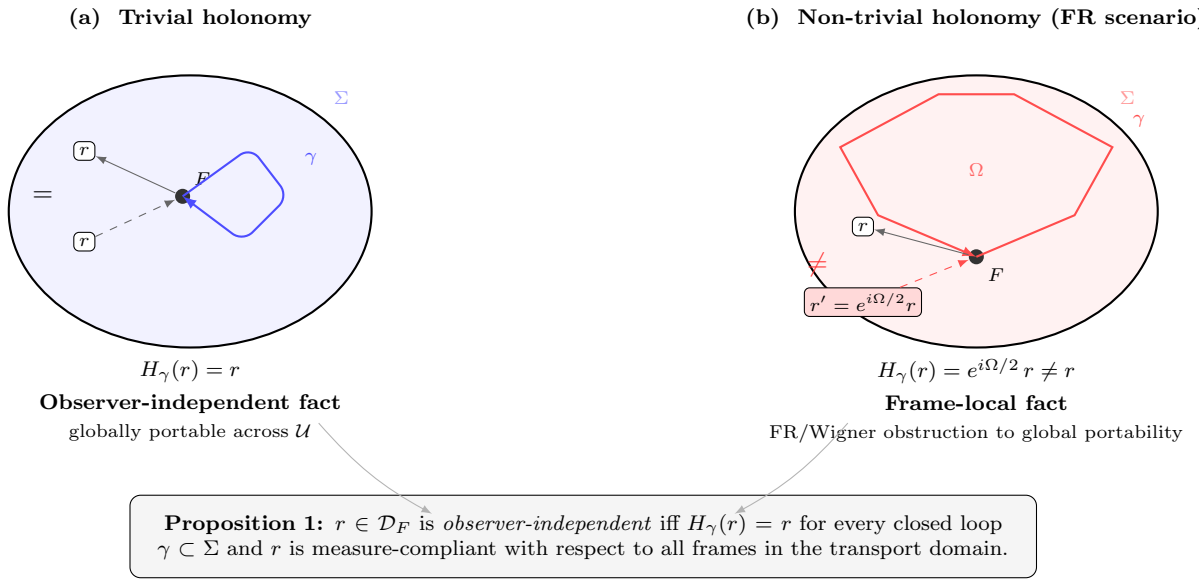

FIG. 2. Holonomy criterion for observer-independent facts (Proposition 1). **(a)** A contractible loop γ in coherence space Σ returns the transported record r unchanged: $H_\gamma(r) = r$. The record is observer-independent—globally portable across all frames in the admissible domain. **(b)** A large loop (the Frauchiger–Renner six-step agent-comparison loop is one instance) encloses nontrivial solid angle Ω in Σ and returns a holonomy-rotated record $r' = e^{i\Omega/2} r \neq r$. The record is frame-local, not globally portable. Apparent contradictions in FR and Wigner's friend scenarios arise when frame-local facts are transported outside their compliance domain as though holonomy were trivial. The antisymmetric part Ω_{AB} of the tensor field T_{AB} (Eq. (7)) is the infinitesimal obstruction: $\Omega_{AB} \neq 0$ makes nontrivial holonomy possible.

E. Frame Invariants

Theorem 1 (Frame Invariants)**.** *The following are invariant under all frame transformations:*

1. *Total probability:* $\sum_i \mu_F(i) = 1$
2. *Born measures of alternatives:* $\mu_F(i) = \mu_{F'}(\pi(i))$ *(for the measure* $\mu_F(i) = |c_i|^2$ *derived in Theorem 2; this item is stated here for completeness and verified in §IV)*
3. *Inner products:* $|\langle\psi|\phi\rangle|^2$ *is frame-independent*
4. *Eigenvalue spectra of observables*
5. *Entanglement entropy* $S(\rho_A)$ *for bipartite systems*

What is *not* invariant: which observables have definite values, whether a state appears coherent or decohered, and the form of the density matrix.

IV. THE BORN RULE AS FRAME-INVARIANT MEASURE

The derivation that follows is Gleason-like in structure but replaces noncontextuality with frame-invariance: the symmetry is not "the same observable measured by different projectors yields the same result" but rather "the same physical situation described in different coherence frames yields the same probabilities."

A. Statement

The key symmetry underlying the derivation is invariance under refinement of alternatives. If two descriptions differ only by coupling to an ancilla that refines a single alternative into multiple orthogonal sub-alternatives of equal amplitude, then no physical probability assignment can distinguish between them. Such refinements are routine in quantum information and correspond operationally to increasing measurement resolution without altering the underlying physical situation. A probability measure that depended on whether or not such a refinement had been performed would be ill-defined, as it would assign different probabilities to physically equivalent descriptions. We formalize this requirement as invariance under frame splitting.

The ancilla splitting construction in Proposition 2 follows the fine-graining technique used in Zurek's envariance derivation [5] and the Deutsch–Wallace approach [6, 7]. The novelty of the present derivation lies not in the construction itself but in its role: rather than establishing envariance or decision-theoretic equivalence, the splitting grounds *frame-invariance on* Σ as the operative symmetry—a geometric condition that holds for $\dim \mathcal{H} \geq 2$ and requires no physical environment or branching ontology.

Theorem 2 (Born Rule from Frame Invariance)**.** *Let* Σ *be the coherence space for* $\mathcal{H}$ *with* $\dim \geq 2$*. Let* μ *be a measure on the alternatives* $\{|i\rangle_F\}$ *satisfying:*

(A1): ***Frame invariance****: μ is invariant under coherence relabelings* (9) *and frame splittings* (15)*.*

(A2): ***Additivity****: μ is additive on mutually orthogonal alternatives.*

(A3): ***Frame locality:*** *$\mu_F(i)$ depends only on $|\psi\rangle_F$ and $|i\rangle_F$—not on the history of frame transformations leading to F, nor on probability assignments in other frames.*

This is a locality assumption on Σ*, analogous to the independence of probability assignments from distant reference frames in relativistic theories—a postulate reflecting the physical structure of coherence space, not a derived result. It excludes contextual hidden-variable models in which the probability of outcome i depends on which other measurements are simultaneously possible.*

The physical justification is two-part. First, if $\mu_F(i)$ depended on the history of frame transformations leading to F, then the same coherence frame reached by different preparation paths would yield different predictions—making "coherence frame" an ill-defined concept whose content depends on unobservable history. Second, (A3) is strictly weaker than Gleasonś non-contextuality: it requires only path-independence within Σ, not global independence across all POVMs. We regard this weaker form as uncontroversial given the operational definition of a coherence frame as a physically distinct descriptive perspective (§II). Readers who wish to treat (A3) as an assumption rather than a motivation may do so without affecting the formal derivation.

(A4): ***Continuity****: $\mu_F(i)$ is continuous in the coefficients c_i.*

Then $\mu_F(i) = |c_i|^2$, where $|\psi\rangle_F = \sum_i c_i |i\rangle_F$.

B. Derivation

Step 1: Phase independence. By (A1), μ is invariant under coherence relabelings (9), including arbitrary phase rotations θ_i. Hence $\mu_F(i)$ depends on c_i only through $|c_i|$.

Step 2: Equal amplitudes. For $|\psi\rangle_F = N^{-1/2}\sum_{k=1}^{N}|k\rangle_F$, permutation invariance gives $\mu_F(k) = 1/N = |c_k|^2$ for all k.

Step 3: Rational amplitudes. For $|\psi\rangle_F = \sqrt{m/(m+n)}\,|0\rangle_F + \sqrt{n/(m+n)}\,|1\rangle_F$, construct a frame splitting (Prop. 2) producing an equal-amplitude state of $m+n$ terms. By Step 2 and additivity (A2):

$$\mu_F(0) = \frac{m}{m+n} = |c_0|^2, \quad \mu_F(1) = \frac{n}{m+n} = |c_1|^2. \quad (18)$$

Step 4: General amplitudes. Approximate $|c_i|^2$ by rationals $r_i^{(n)}$ with $|r_i^{(n)} - |c_i|^2| < 1/n$. By Step 3, $\mu^{(n)}(i) = r_i^{(n)}$. By continuity (A4), $\mu_F(i) = \lim_{n\to\infty}\mu^{(n)}(i) = |c_i|^2$. □

C. Comparison with Previous Derivations

Table III summarizes the relationship to prior work. The structural parallel with Zurek's envariance [5] warrants explicit comparison. Both derivations use a splitting step to reduce general amplitudes to equal amplitudes. However, they differ in three respects: (i) Zurek's envariance requires a physical environment and invokes swap symmetry of the system–environment state, which critics [4] have argued presupposes the Born rule; our splitting (Prop. 2) is a mathematical theorem about Hilbert space embeddings, requiring no environment. (ii) Envariance acts on a bipartite state $|\psi\rangle_{SE}$; frame invariance acts on the geometric structure of Σ itself, making it independent of any particular physical setup.

A precise comparison with Gleason's theorem is warranted. Gleason's non-contextuality (NC)—the structural assumption shared with the Kochen–Specker theorem [26]—is a *global* condition: a projector P must yield the same probability regardless of which resolution of the identity it belongs to—across *all* POVMs simultaneously. Axiom (A1) requires something strictly weaker: that isometric embeddings of the form in Proposition 2—equal-amplitude frame splittings via ancilla coupling—preserve probability assignments. This is a *local geometric* condition on Σ, placing no constraint on probabilities for projectors in unrelated measurement contexts. The logical relationship is: NC $\Rightarrow$ (A1), but (A1) $\not\Rightarrow$ NC. We note additionally that Gleason's theorem requires $\dim\mathcal{H} \geq 3$; the present derivation holds for $\dim \geq 2$ because the splitting argument operates on the ancilla-extended space, bypassing the qubit obstruction.

(iii) The coherence-relativity derivation makes the *geometric origin* of the symmetry manifest: frame invariance is the requirement that physics does not depend on which coherence frame is used for description, just as Lorentz invariance requires independence of inertial frame.

The present derivation establishes $\mu_F(i) = |c_i|^2$ for pure states via (A1)–(A4). Extension to mixed states follows by purification: any mixed state ρ on $\mathcal{H}_S$ is the partial trace of a pure state $|\Psi\rangle$ on $\mathcal{H}_S \otimes \mathcal{H}_R$, and the pure-state result recovers $\mu_F(i) = \mathrm{Tr}(\rho|i\rangle\langle i|)$ by tracing out $\mathcal{H}_R$. Purifications are not unique—different choices of $\mathcal{H}_R$ and $|\Psi\rangle$ purify the same ρ—but the Born measure $\mu_F(i) = \mathrm{Tr}(\rho|i\rangle\langle i|)$ depends only on ρ and $|i\rangle_F$, not on the choice of purification, by the cyclicity of the partial trace. The extension is therefore well-defined. The circularity objection applying to envariance—which requires a physical environment and swap symmetry—does not apply here: frame splittings (Prop. 2) are mathematical theorems about Hilbert space embeddings requiring no environment.

TABLE III. Comparison of Born rule derivations.

Derivation	Symmetry Used	Physical Content	Assumes Born?	Result
Gleason (1957)	Non-contextuality	Mathematical ($\mathcal{H}$)	No	$\mu = \mathrm{Tr}(\rho P)$
Zurek (2005)	Envariance	Entanglement + environment	Debated	$\mu = \lvert c_i \rvert^2$
Deutsch–Wallace	Rational equivalence	Many-worlds ontology	Debated	$\mu = \lvert c_i \rvert^2$
Masanes et al. [27]	Operational axioms	Compositional structure	No	Born rule unique
This work	**Frame invariance**	**Geometric** (Σ, T_{AB})	**No**	$\boldsymbol{\mu = \lvert c_i \rvert^2}$

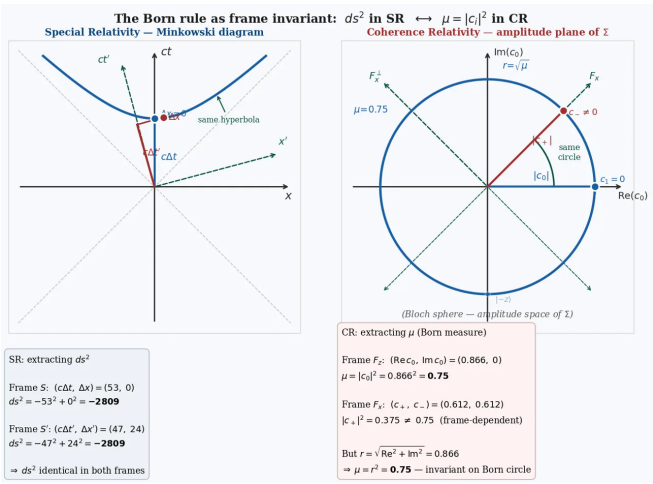

FIG. 3. The Born rule as frame invariant: ds^2 in SR $\leftrightarrow$ $\mu = |c_i|^2$ in HCR. **Left (SR):** Minkowski diagram showing two frames S, S'; the interval ds^2 is identical in both. **Right (HCR):** Amplitude plane of Σ (Bloch sphere cross-section); the Born measure $\mu = r^2 = |c_0|^2$ is invariant under frame rotations, while individual components $|c_+|^2$ are frame-dependent.

D. Physical Interpretation

The Born rule is not a postulate, a frequency statement, or a consequence of branching. It is the **unique frame-invariant probability assignment** on Σ—the quantum analog of the invariant interval ds^2 in special relativity (Fig. 3).

V. PARADOX RESOLUTIONS AND WORKED EXAMPLES

Coherence relativity resolves quantum paradoxes by reframing them as frame-mismatch problems. In each case, the apparent paradox arises from transporting a Born weight from a frame F with a fully populated accessible sector to a frame F' with a partially populated accessible sector—due to degrees of freedom having been traced into the bath or never entering the accessible sector—without applying the appropriate frame transformation. The transported weight incorrectly inherits the normalization structure of the source frame rather than the target frame's accessible sector. The resolution is always the same: apply the frame transformation, compute the Born measure in the target frame, and recognize that the two frames give different but mutually consistent descriptions of a single reality $\mathcal{R}$.

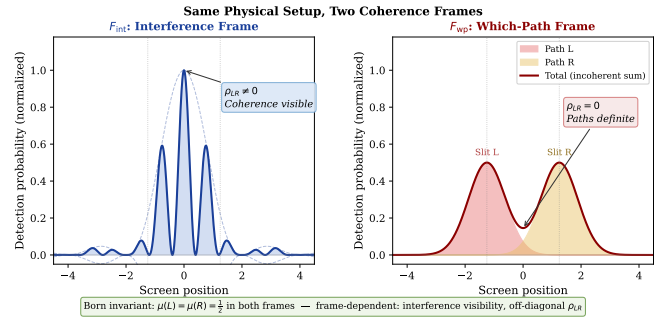

FIG. 4. Double-slit experiment in two coherence frames. Left: interference frame F_{int} shows the characteristic fringe pattern. Right: which-path frame F_{wp} resolves individual paths with no interference. The Born measure $\mu(L) = \mu(R) = 1/2$ is invariant.

A. Double-Slit Experiment

Consider the double-slit with two frames (Fig. 4):

- F_{int}: $|\psi\rangle_{F_{\mathrm{int}}} = \frac{1}{\sqrt{2}}(|L\rangle + |R\rangle)$. Interference visible.
- F_{wp}: $|\psi\rangle_{F_{\mathrm{wp}}} = \frac{1}{\sqrt{2}}(|L\rangle\,|W_L\rangle + |R\rangle\,|W_R\rangle)$. Paths definite.

The transformation $U_{F_{\mathrm{int}} \to F_{\mathrm{wp}}}$ is realized by detector coupling (Fig. 5). Frame invariants: $\mu(L) = \mu(R) = 1/2$ in both frames. Frame-dependent: visibility of interference fringes, off-diagonal density matrix elements.

B. Delayed-Choice and Quantum Eraser

Recent delayed-choice and quantum-eraser experiments sharpen the timing and control of the original Wheeler and Scully–Drühl proposals, but they do not change the basic puzzle. Quantum-computing realizations of the eraser [28] and dual-selection delayed-choice interferometry [29] make clear that the issue is not whether the future reaches back into the past, but which

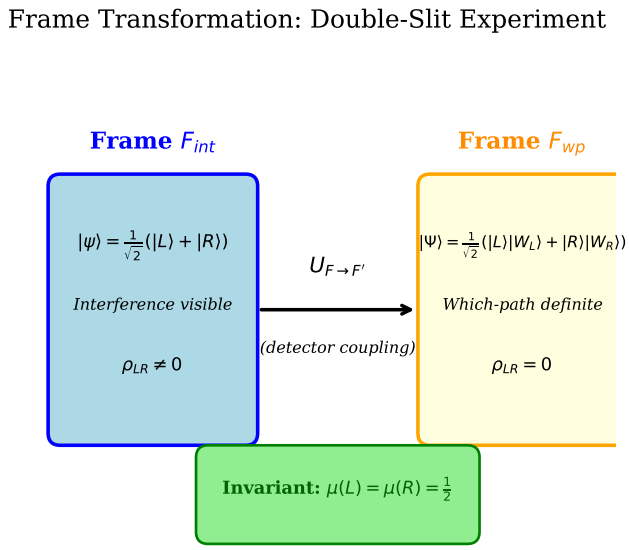

FIG. 5. Operator-level frame transformation $U_{F_{\rm int}\to F_{\rm wp}}$ for the double-slit experiment. Left (**blue**): interference frame $F_{\rm int}$ in which the system is a coherent superposition $|\psi\rangle = \frac{1}{\sqrt{2}}(|L\rangle + |R\rangle)$; interference is visible and $\rho_{LR} \neq 0$. Right (**orange**): which-path frame $F_{\rm wp}$ after detector coupling; paths are definite and $\rho_{LR} = 0$. The frame transformation $U_{F_{\rm int}\to F_{\rm wp}}$ is unitary, preserving the Born measure $\mu(L) = \mu(R) = \frac{1}{2}$ (green strip), while changing which observables have definite values. Compare Fig. 4 for the experimental geometry.

comparison frame is used to describe the completed experiment.

In HCR, the late choice does not rewrite an already fixed past. It selects the final comparison frame and therefore the record algebra in which the experiment is described. If the closing frame preserves which-path records, the experiment ends in a definite-path description. If the closing frame erases the distinguishing records, it ends in an interference-compatible description. What changes is not an earlier fact, but whether the earlier interaction had already become a Born-compliant, observer-independent record across the relevant transport loop.

This is the temporal point developed in Sec. VII: an event can be earlier in coordinate time without yet belonging to the HCR-past. Delayed-choice experiments therefore do not force retrocausality; they display frame-relative fact-establishment. The later intervention completes the trajectory through Σ and determines which facts are jointly portable, rather than altering a past that was already fixed.

C. Wigner's Friend

Wigner's friend [30] performs a measurement inside a laboratory (frame $F_{\rm friend}$: state has collapsed to $|\uparrow\rangle$ or $|\downarrow\rangle$). Wigner, outside the lab, describes the joint system in $F_{\rm Wigner}$: the friend–system composite remains in superposition. In HCR, both descriptions are correct *in their respective frames*. There is no contradiction, just as there is no contradiction between observers disagreeing about simultaneity in SR.

D. Frauchiger–Renner and Observer-Independent Facts

Frauchiger and Renner [31] sharpen Wigner's friend by allowing agents to use quantum theory to reason about one another across nested laboratories. Proietti et al. [32] realized a related scenario experimentally and observed a violation of the associated Bell-Wigner inequality under auxiliary assumptions of locality and free choice. Brukner [33] formalized the corresponding no-go theorem for observer-independent facts. These results are often read as forcing a choice among three claims: the universal validity of quantum theory, the existence of single definite outcomes, and the consistency of agents' conclusions. In HCR, the issue is more specific. The relevant comparisons lie within the unitary transport structure of Sec. III C; more general non-unitary transport is left to Paper 2.

These scenarios form closed loops of comparison in coherence space. A record $r \in \mathcal{D}_{F_{\rm friend}}$ is definite for the friend, while an outside super-observer transports the same laboratory to a frame in which the joint lab remains coherent and is checked by an interference-type measurement. The contradiction appears only if one assumes that the friend's record and the super-observer's result can both be carried unchanged around the full loop as observer-independent facts.

Proposition 1 says that such transport is allowed only when the comparison domain is compliant and the holonomy is trivial, $H_\gamma(r) = r$, on every closed loop γ (Fig. 2). Frauchiger–Renner fails exactly at this point. Each agent's inference is valid inside the frame in which its record is definite, but inconsistency appears when frame-local facts are moved around the loop as if holonomy were trivial. The problem is therefore not an inconsistency in quantum theory itself, but an obstruction to global fact assignment.

The same diagnosis applies to the Proietti experiment. The friends' facts A_0, B_0 and the super-observers' interference-type facts A_1, B_1 do not fit into a single observer-independent record algebra. In HCR, that is exactly what one expects when the relevant loop in coherence space carries nontrivial holonomy.

HCR therefore dissolves, rather than merely chooses a horn of, the Frauchiger–Renner trilemma. Universal unitary description may hold, and single outcomes may exist within the frames where they are definite, without implying that every local fact is globally portable around the full Wigner loop. What fails is the extra assumption of flat, observer-independent transport. Ex-

plicit holonomy calculations for Frauchiger–Renner-type and Proietti-type loops are deferred to Paper 2C.

E. Quantum Cheshire Cat

Recent work on the quantum Cheshire cat increasingly treats the effect as a problem about contextuality and comparison procedures, not literal separation of an object from its properties [34, 35]. The experiments use pre- and post-selection, together with weak access to different observables, to produce signatures that invite an unusual metaphysical reading.

In HCR, the problem appears when path-claims and property-claims are treated as if they belonged to one observer-independent record algebra. Proposition 1 says that this is justified only when the relevant records satisfy compliance and trivial holonomy. If the claims come from different comparison domains, there is no reason to expect unrestricted joint portability.

HCR therefore accepts the empirical signatures while rejecting the literal picture of a property floating free of its carrier. What the experiments show is a limit on how different frame-local claims can be combined, not the existence of detached properties in a frame-invariant sense.

F. Weak Measurements and Objective Realism

Recent weak-measurement arguments against objective realism are best read as challenges to naive global realism, not to reality itself. Results such as the independent weak-measurement violation of objective realism reported in Ref. [36] question whether all queried quantities can be assigned simultaneously portable definite values within one comparison scheme.

In HCR, Proposition 1 provides the test. A weak value or postselected quantity does not become an observer-independent fact merely because it enters a successful inference. It becomes global only if it is instantiated in definite records and transported around the relevant loop with trivial holonomy, $H_\gamma(r) = r$. Otherwise it remains frame-local: meaningful within the protocol that defines it, but not automatically promotable to a universal value assignment.

This connects directly to the latent-versus-instantiated Born discussion of Sec. VII. The Born measure is present throughout the protocol, including in coherent and weakly probed regimes. What changes from frame to frame is whether a stable record has formed that turns that measure into an actual fact about a definite outcome. On this reading, weak-measurement paradoxes expose the limits of global fact portability, not a failure of reality itself.

G. EPR and Bell Nonlocality

Einstein, Podolsky, and Rosen [37] argued that quantum mechanics is incomplete based on correlations between separated systems. For an entangled pair in the singlet state, Alice's and Bob's local frames each see mixed states. The global frame sees entanglement. Bell inequality violation is a frame-invariant fact (it depends only on $|\langle\psi|\phi\rangle|^2$ terms). The correlations are not "caused" by collapse but are geometric properties of the entangled state visible from any frame.

H. Schrödinger's Cat

The cat-in-a-box scenario involves two frames: F_{outside} (cat in superposition) and F_{inside} (cat alive or dead). Opening the box is a frame transformation $U_{F_{\text{outside}}\to F_{\text{inside}}}$, not a physical collapse. The cat's definite state exists in F_{inside} at all times; it simply isn't accessible from F_{outside} without the frame transformation.

This is quantitative, not merely verbal. Joos–Zeh estimates and their later experimental confirmation show that macroscopic record states are manufactured collision by collision on extremely short, environment-dependent timescales; absent such monitoring, even macroscopic orientational states such as Hyperion's would remain coherently delocalized rather than classically definite [9, 16, 17].

I. Quantum Zeno Effect

Frequent measurements prevent a system from evolving—how can observation stop dynamics? The Zeno effect admits several explanations (projection [38]; Zeno subspace confinement [39]; continuous environmental monitoring). HCR provides a unified geometric language: each explanation becomes a distinct physical claim about what happens in Σ during measurement, and each leads to distinguishable predictions (a full four-explanation comparative framework with experimentally independent discriminating axes is developed in Paper 2C).

The HCR-specific contribution is the D_f completion interpretation: each strong projective measurement completes the quantum-to-classical transition, fully populating the diagonal density matrix in the pointer basis ($\rho_{\text{pointer}} = \sum_i p_i|i\rangle\langle i|$, all off-diagonals zero). The Zeno regime corresponds to $\tau_{\text{meas}} < \tau_{\text{transform}}$ (Eq. 26): each measurement completes D_f before Hamiltonian evolution can repopulate coherences. The anti-Zeno regime corresponds to $\tau_{\text{meas}} > \tau_{\text{transform}}$: D_f is not fully populated between steps, coherences survive, and the system evolves into uncovered regions of Σ.

This completion picture aligns naturally with the measurement-induced phase transition (MIPT) litera-

ture [40–43]. In that language, the threshold $\tau_{\text{meas}} \approx \tau_{\text{transform}}$ plays the role of a critical monitoring scale: below it the stationary distribution on Σ remains spread over coherent frames, while above it weight concentrates near pointer frames and classicalization stabilizes. On the present view, the MIPT is the many-body manifestation of the same D_f-completion threshold, with the coherence-relativity quasipotential developing the non-analytic structure characteristic of the transition.

Regardless of which Zeno mechanism is correct, the paradox follows the general pattern: the naive observer imports the partially-populated free-evolution dynamics (quantum evolution ongoing, coherences present) into the measurement frame (where measurement has at least partially populated D_f) without applying the frame transformation.

This yields a seventh experimentally testable signature: the Zeno/anti-Zeno crossover occurs at $\tau_{\text{meas}} \approx \tau_{\text{transform}}$. Standard QM predicts the crossover from spectral-density arguments [44]; HCR predicts it from the D_f completion condition. The cleanest HCR-specific test: two measurements at the same rate but in different bases should give different crossover behavior if one basis matches the pointer basis and the other does not—a pointer-basis dependence not predicted by standard Zeno treatments.

J. Sleeping Beauty and Self-Locating Uncertainty

The Sleeping Beauty problem [45]—where a fair coin determines whether an amnestic subject is awakened once or twice, and she must assign credence to Heads upon waking—has generated over 100 papers [46] without resolution. The "halfer" answer (1/2) preserves conditionalization but violates indifference; the "thirder" answer (1/3) respects indifference but violates the principle that credence should not shift without new information. Sebens and Carroll [47] proved this paradox has the exact same logical structure as the central open problem in Everettian quantum mechanics: how to derive the Born rule when all branches are realized.

In HCR, the halfer/thirder debate is a frame-mismatch error. Beauty before sleep is in frame F_{pre}, where her state $|\psi\rangle_{F_{\text{pre}}} = (|H\rangle + |T\rangle)/\sqrt{2}$ assigns credence 1/2 to Heads via the Born measure. The amnestic drug physically implements a Hamiltonian coupling that transfers temporal self-location information from Beauty's accessible system into an inaccessible biochemical bath—structurally identical to measurement-induced decoherence. This is a standard frame transformation to F_{wake}, where the accessible sector contains three centered alternatives $\{H_{\text{Mon}}, T_{\text{Mon}}, T_{\text{Tue}}\}$.

By (A3), the Born measure in F_{wake} depends only on $|\psi\rangle_{F_{\text{wake}}}$ and $|i\rangle_{F_{\text{wake}}}$—information outside the frame cannot enter the measure. The coin outcome, never received by Beauty, is excluded by (A3). By (A1), the three permutation-symmetric alternatives in the accessible sector receive equal weight: $\mu_{F_{\text{wake}}}(H_{\text{Mon}}) = \mu_{F_{\text{wake}}}(T_{\text{Mon}}) = \mu_{F_{\text{wake}}}(T_{\text{Tue}}) = 1/3$. The halfer's 1/2 is correct in F_{pre}; the thirder's 1/3 is correct in F_{wake}. The paradox arises from demanding a single frame-independent credence—the coherence-frame analog of demanding absolute simultaneity in SR.

This suggests a sharpened preferred-basis criterion: a basis is preferred iff self-locating propositions can be consistently assigned truth values within it. Decoherence then selects not merely stable pointer states, but the basis in which questions of the form "which outcome am I in?" are well-posed. In the Sleeping Beauty protocol, the centered alternatives $\{H_{\text{Mon}}, T_{\text{Mon}}, T_{\text{Tue}}\}$ are exactly the alternatives that remain in Beauty's accessible sector at F_{wake}, so they constitute the preferred basis for her post-amnesia credence assignment.

Beauty's amnesia is therefore the destruction of a temporal record, not the creation of new information. The transition $F_{\text{pre}} \to F_{\text{wake}}$ removes the "which awakening" record from the accessible sector of Σ, increasing self-locating uncertainty by decohering a temporal degree of freedom into the bath. In this sense the protocol fits the temporal analysis of Sec. VII: in coherence relativity, what counts as past is the structure of presently available records, so erasing a temporal record reopens centered uncertainty without changing the underlying coin history.

Sebens and Carroll's epistemic separability principle—that changes to the environment alone do not affect local probability assignments—is compatible with Born measure invariance (A1) and follows from (A1) applied to frame transformations that act only on the bath sector of Σ. The structural identity between Sleeping Beauty and Everettian branch probability means that HCR's resolution automatically provides a geometric derivation of the Born rule in the many-worlds setting, without requiring the decision-theoretic apparatus of Deutsch–Wallace [6, 7] or the envariance argument of Zurek [5].

VI. DISCUSSION

A. Resolution of the Measurement Problem

HCR offers a relational resolution of the measurement problem: "collapse" is a frame transformation, not a dynamical process. The Born rule is not postulated but derived as the unique frame-invariant measure. Definite outcomes exist in appropriate coherence frames; superposition exists in others. There is no need to explain *when* or *how* collapse occurs, because collapse is not a physical event—it is a change of description. Crucially, no frame ever loses coherence: what changes is the geometric relationship between alternatives, as distinguishability becomes redundantly encoded in the environment. Coherence is not destroyed but *re-described* from the perspective of a different frame.

A causal-direction caveat is important here. The joint

space $\mathcal{M} \times \Sigma$ should not be read as if spacetime drives coherence space, or vice versa, at the ontological level. Within the constituted arena the dynamics are bidirectional: T_{AB} couples $\mathcal{M}$ and Σ simultaneously, and when we say that "the environment drives decoherence" the environment is itself already a projection of the deeper structure into $\mathcal{M}$-coordinates. The ontological direction is constitutive rather than dynamical, schematically $\mathcal{H} \rightsquigarrow \Sigma \rightsquigarrow \mathcal{M}$.

B. Frame-Relative Pastness and Factual Closure

Coherence relativity suggests a frame-relative account of when facts become settled. In special relativity, simultaneity is not intrinsic to events alone; it depends on the inertial frame used to compare them. In HCR, pastness works similarly. An event counts as past for a frame F when it is represented in F by a definite record $r \in \mathcal{D}_F$. It becomes an observer-independent past fact only when the compliance and holonomy conditions of Proposition 1 are met. The future consists of alternatives still coherent in the relevant frame, and the present is the stage at which those alternatives are being turned into definite records. This does not add a second time parameter. It says instead that fact-closure on Σ need not track coordinate-time ordering on $\mathcal{M}$ exactly.

This helps with delayed-choice and other retrocausal puzzles. An event can happen earlier in coordinate time and still fail to count as part of the observer-independent past if the relevant transport loop has not yet closed. In that case it remains, for some frames, in the present process of fact-establishment. What is already past for the friend can still be present for Wigner. The apparent retrocausality comes from treating a frame-local description as if it were already frame-invariant. In this sense, SR and HCR express the same general lesson: temporal relations depend on the frame at the level where they are defined. The extension of this parallel to the full geometry of $\mathcal{M} \times \Sigma$ is deferred to Paper 2.

C. Quasipotential Landscape

The tensor field T_{AB} (Sec. II D) generates a quasipotential landscape on Σ via an action principle. For a path $\gamma : \sigma \mapsto x(\sigma)$ through coherence space with effective drift $F^A(x)$ (driven by system–environment coupling) and noise scale D, define the action

$$\mathcal{S}[\gamma] = \frac{1}{4D} \int_0^1 \left(\dot{x}^A - F^A\right) G_{AB} \left(\dot{x}^B - F^B\right) d\sigma, \qquad (19)$$

where $\dot{x}^A \equiv dx^A/d\sigma$ and $\sigma \in [0,1]$ is a path parameter distinct from the geometric arc length $s(\lambda) = \int_0^\lambda \sqrt{G_{\lambda\lambda}}\, d\lambda'$ introduced below.

This is the Freidlin–Wentzell large-deviation functional [48], which defines the minimum-cost path for stochastic excursions away from an attractor. The quasipotential is $V(x) = \inf_\gamma \mathcal{S}[\gamma]$, where the infimum is over paths from a reference frame to x (Fig. 7). Minima of V correspond to metastable classical frames (pointer states), saddle points determine transition rates, and the stationary distribution concentrates as $p(x) \propto \exp(-V(x)/\epsilon)$ for small effective noise ϵ. This provides a geometric explanation for einselection: environment-induced superselection is the dynamical relaxation to V-minima. In the purely geometric limit (zero drift $F^A = 0$, zero noise $D \to 0$), the stochastic quasipotential $V_{\text{stoch}}(x) = \inf_\gamma \mathcal{S}[\gamma]$ reduces to the *geometric quasipotential* $V(\lambda) = d_\Sigma(\lambda) = \int_\lambda^1 \sqrt{G_{\lambda\lambda}}\, d\lambda'$—the Fubini–Study distance from λ to the classical frame. It is this geometric quasipotential that we compute and plot in Figs. 7 and 8; the full stochastic V_{stoch} requires the explicit drift F^A, which is the primary object of Paper 2.

The novel application is the identification of Σ as the arena and G_{AB} as the cost tensor—a construction with no prior precedent in the quantum decoherence literature. We conjecture that F^A, when derived explicitly from the system–environment interaction Hamiltonian H_{SE}, drives coherence-space trajectories to einselected pointer-state frames; this is consistent with Zurek's einselection picture [4] and is the primary object of Paper 2. Here F^A enters as a phenomenological drift term whose form is constrained by the requirement that V-minima reproduce known pointer states in tractable models.

The accumulated Fubini–Study length $s(\lambda) = \int_0^\lambda \sqrt{G_{\lambda\lambda}}\, d\lambda'$ quantifies the total geometric distance traversed from the coherent frame, so that $V(\lambda) = s(1) - s(\lambda)$ is the geometric quasipotential (distance remaining to the pointer-state minimum), valid in the zero-drift limit. For the multi-mode model ($N = 10$), the total length is $s(1) \approx 1.46$; for $N = 1$, $s(1) = \pi/(2\sqrt{2}) \approx 1.11$. Under the EGY parameterization, $s(\lambda)$ grows from 0 at $\lambda = 0$ and accelerates near $\lambda \to 1$ (where $G_{\lambda\lambda}$ diverges). The quasipotential $V(\lambda) = s(1) - s(\lambda)$ therefore decreases steeply near $\lambda \to 1$, reflecting the increasing geometric cost of maintaining classical records. This is structurally analogous to a potential landscape where the classical fixed point at $\lambda = 1$ sits at the bottom of a well with walls that steepen approaching the pointer basis—the system must traverse the full geometric length $s(1)$ to fully establish classical distinguishability.

We refer to $ds/d\lambda = \sqrt{G_{\lambda\lambda}}$ as the *geometric entropy production rate* and to $s(\lambda)$ as the *accumulated geometric entropy*—the total information-geometric arc length traversed from the coherent frame. These quantities are plotted in Fig. 6 and discussed further in Remark 8.

Remark 8 (Frame-dependence of thermodynamic quantities)**.** *The von Neumann entropy S_{vN}, the free energy $\mathcal{F} = \langle H \rangle - TS_{\text{vN}}$, and the sign of environmental heat flow during decoherence are all* frame-dependent *quantities in the sense of HCR: they change under coherence-frame transformations because they depend on where the system–bath boundary is drawn, and that boundary choice*

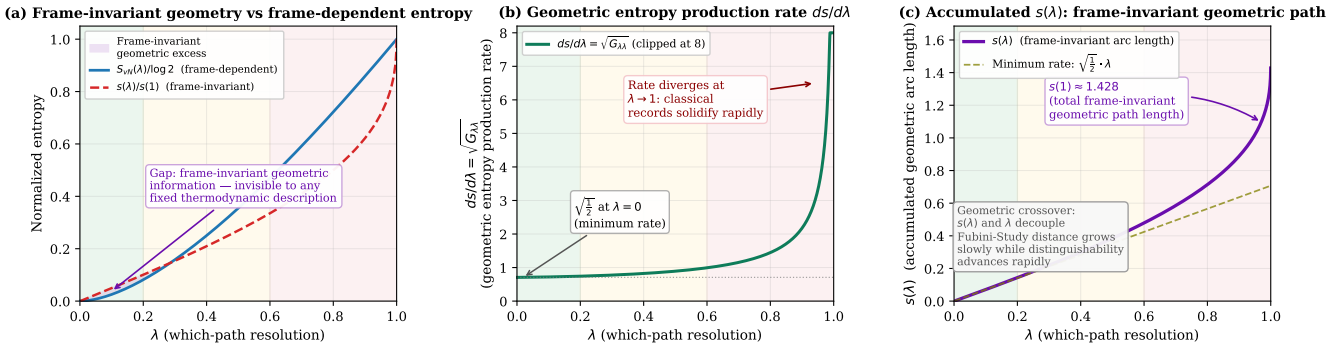

FIG. 6. Geometric entropy in coherence space Σ. **(a)** The Fubini–Study arc length $s(\lambda)/s(1)$ (red dashed, frame-invariant) and von Neumann entropy $S_{\mathrm{vN}}/\log 2$ (blue solid, frame-dependent) both run from 0 to 1 but differ structurally: S_{vN} changes when the system–bath boundary is redrawn (a coherence-frame transformation), while $s(\lambda)$ does not. The shaded gap $\Delta(\lambda)$ is frame-invariant geometric information invisible to any fixed thermodynamic description (Remark 8). **(b)** Geometric entropy production rate $ds/d\lambda = \sqrt{G_{\lambda\lambda}}$: minimal at $\lambda = 0$, diverging as $\lambda \to 1$. **(c)** Accumulated $s(\lambda)$: Fubini–Study distance and distinguishability decouple near the quantum-classical crossover ($\lambda \in [0.2, 0.6]$).

is precisely a choice of coherence frame. Concretely, the heat dissipated into the environment during pure decoherence can be positive, zero, or negative depending on the bath model and coupling [49]: the same physical process R on $\mathcal{M}\times\Sigma$ admits thermodynamic descriptions with any sign of environmental energy exchange, depending on the frame used to partition system from environment.

The Fubini–Study arc length $s(\lambda)$, by contrast, is a property of the trajectory on Σ that is independent *of this partition. It is therefore frame-invariant in the same sense as the Born measure $\mu_F(i) = |c_i|^2$: both are quantities on Σ that do not change when the system–bath boundary is redrawn. The gap*

$$\Delta(\lambda) \;=\; \frac{s(\lambda)}{s(1)} \;-\; \frac{S_{\mathrm{vN}}(\lambda)}{\log 2} \tag{20}$$

visible in Fig. 6(a) is therefore not merely a quantitative difference between two entropy measures: it is the frame-invariant geometric information that is invisible to any fixed thermodynamic description of the decoherence process. *A thermodynamic observer who fixes a system–bath boundary sees S_{vN}; the geometry of Σ knows $s(\lambda)$ regardless of where that boundary is drawn. This reinforces the central claim of HCR: physical reality is more completely described on $\mathcal{M}\times\Sigma$ than on $\mathcal{H}$ alone, and the geometric structure of Σ captures content that is invisible to the density-matrix description.*

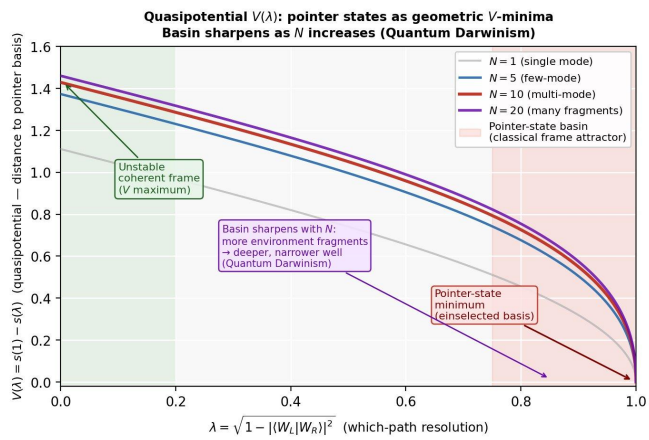

FIG. 7. Quasipotential $V(\lambda) = s(1) - s(\lambda)$ on coherence space for $N = 1, 5, 10, 20$ environment fragments (EGY parameterization). $V(\lambda)$ is maximum at $\lambda = 0$ (unstable coherent frame) and zero at $\lambda = 1$ (pointer-state minimum, einselected basis). The basin sharpens and deepens as N increases: each additional environment fragment independently constrains the preferred coherence frame, geometrically narrowing the V-minimum—the geometric signature of Quantum Darwinism. The shaded region marks the classical frame attractor ($\lambda > 0.75$).

D. Initiation State and Coherence-to-Pointer Transition

A minimal initiation model is given by an initially coherent system–environment state

$$|\Psi_0\rangle = \sum_i c_i \, |i\rangle_S \otimes |E_0\rangle, \tag{21}$$

with interaction map

$$U_{SE}: \ |i\rangle_S|E_0\rangle \mapsto |i\rangle_S|E_i\rangle. \tag{22}$$

Define the decoherence distinguishability coordinate

$$\lambda_D := 1 - \left|\langle E_i|E_j\rangle\right| \quad (i \neq j), \tag{23}$$

with $\lambda_D \in [0,1]$, where i,j label distinct pointer branches. In this paper, the geometric order parameter λ is taken to be a monotone calibration of λ_D; unless otherwise stated, we work in the gauge $\lambda \equiv \lambda_D$. Then $\lambda_D = 0$ corresponds to fully coherent alternatives and $\lambda_D \to 1$ to pointer-resolved records. The reduced state

$$\rho_S(\lambda_D) = \mathrm{Tr}_E[|\Psi(\lambda_D)\rangle\langle\Psi(\lambda_D)|] \tag{24}$$

approaches diagonality in a dynamically stable pointer basis when decohering couplings dominate recohering channels. In this framework, classical spacetime is not assumed at $\lambda_D = 0$; rather, it is an effective description associated with the high-distinguishability sector ($\lambda_D \to 1$) and the induced macroscopic observables. C^1 regularity is adopted here as a transition-level condition on the coherence-to-classical map: the trajectory on Σ is smooth in information-geometric distance, and the induced macroscopic fields are continuous with continuous first derivatives across the quantum-classical crossover. The explicit form of the emergent metric and the embedding geometry are developed in Paper 2.

E. Quantum Darwinism and Objectivity

HCR naturally accommodates quantum Darwinism [50]: when multiple environment fragments independently acquire the same information about the system, they select a preferred coherence frame. In HCR language, the redundant encoding of pointer-state information across environment fragments [51, 52] is consistent with the narrowing of the quasipotential well around the classical frame as the number of monitoring fragments N increases: each additional fragment independently constrains the preferred coherence frame, geometrically sharpening the V-minimum in a manner made precise in Paper 2. In Fig. 7, each value of N corresponds to a distinct multi-mode metric $G^{(N)}_{\lambda\lambda}(\lambda)$ (computed as in Fig. 1 with N collectively coupled environment modes); larger N yields a larger total arc length $s^{(N)}(1)$ and therefore a deeper, narrower V-minimum at $\lambda = 1$. The present paper notes only the structural consistency between the QD redundancy criterion and the geometric quasipotential picture; a quantitative derivation requires the explicit form of F^A from H_{SE}.

F. Testable Predictions

Before stating the predictions, we identify the physical origin of the deviations from standard quantum mechanics. The predictions do *not* arise from modified Hamiltonian dynamics, alterations to the unitary group, or replacement of the Born rule—the full unitary structure of quantum mechanics is preserved on $\mathcal{H}$. The deviations arise instead from the *geometric structure of* Σ: specifically, the Fubini–Study metric $G_{\lambda\lambda}$ and the quasipotential $V(\lambda)$, which standard quantum mechanics does not model. Standard QM implicitly treats all traversals of the decoherence parameter λ as metrically equivalent; Coherence Relativity predicts that physically traversed paths are weighted by $G_{\lambda\lambda}$. We acknowledge that the geometric structure of Σ constitutes additional structure beyond standard QM's formalism; this framework is therefore a minimal *geometric extension* that becomes empirically distinguishable from standard QM precisely through predictions 1–6 below.

Predictions (1) (non-linear visibility) and (5) (transformation timescale) follow directly from $G_{\lambda\lambda}$ via the geodesic integral (Eq. 26); they require no dynamical assumptions beyond the metric. Predictions (2) (hysteresis $\delta\mathcal{V}$), (3) (curvature scaling), (4) (gravitational decoherence modification), and (6) (coherence revival) additionally invoke the stochastic action $\mathcal{S}[\gamma]$ and drift F^A, whose explicit derivation from system–environment coupling is deferred to Paper 2. The order-of-magnitude estimates for these four predictions are motivated by the geometric structure but should be understood as indicative rather than derived. The quantitative falsifiability of predictions (1) and (5) is therefore independent of Paper 2.

Eight experimentally testable signatures distinguish HCR from standard QM:

1. **Non-linear visibility**: the HCR geometric traversal $d_\Sigma(\lambda)/d_\Sigma(0)$ deviates from the standard QM visibility $\mathcal{V} = \sqrt{1-\lambda^2}$ due to $G_{\lambda\lambda}$ curvature (Fig. 8).

2. **Frame-transformation hysteresis**: Erasure paths $\lambda : 0 \to 1 \to 0$ show path-dependent visibility recovery (Fig. 9a).

3. **Coherence-space curvature**: Many-body systems exhibit $G_{\lambda\lambda}$ scaling deviations from mean-field predictions.

4. **Modified gravitational decoherence**: Gravitational contributions to T_{AB} produce mass-dependent frame-transformation rates differing from Diósi–Penrose [14, 15].

5. **Transformation timescales**: Weak measurements probe finite frame-transformation duration, resolvable with current qubit platforms (Fig. 9b).

6. **Coherence revival**: Closed systems show partial coherence recovery forbidden in irreversible-collapse models [12, 53].

7. **Zeno/anti-Zeno crossover**: The crossover timescale is $\tau_{\mathrm{meas}} \approx \tau_{\mathrm{transform}}$ from the D_f completion condition, distinct from the spectral-density

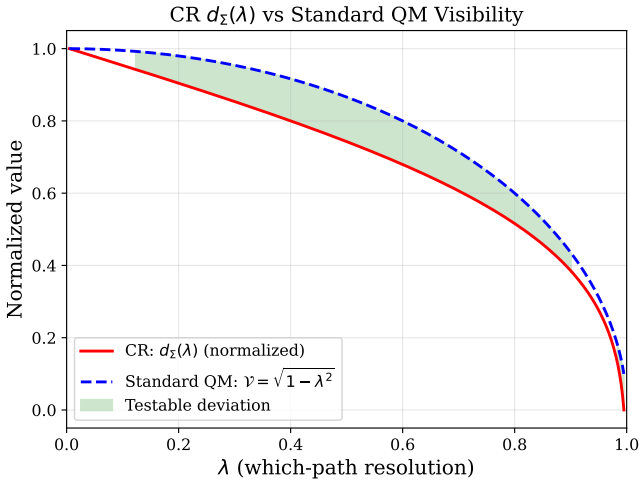

FIG. 8. Comparison of the normalized HCR geometric distance $d_\Sigma(\lambda)/d_\Sigma(0)$ (solid, red) with standard QM visibility $\mathcal{V} = \sqrt{1-\lambda^2}$ (dashed, blue) under the EGY parameterization $\lambda = \sqrt{1 - |\langle W_L|W_R\rangle|^2}$. The HCR curve deviates from the standard QM prediction, reaching a maximum difference $\Delta \approx 0.12$ at $\lambda \approx 0.52$.

prediction [44]. Pointer-basis dependence of the crossover is an HCR-specific signature.

8. **No preferred basis without record constitution**: In decoherence-protected systems that do not sustain stable record-bearing substructures, measurement-like interactions without external record formation should fail to select a unique preferred basis.

Prediction 8 can be operationalized in engineered interferometric platforms by comparing decoherence-protected superconducting or trapped-ion devices with and without an explicit record-forming coupling. HCR predicts that preferred-basis selection tracks the constitution of stable record-bearing structure, not merely the presence of an abstract measurement Hamiltonian.

1. *Quantitative estimates*

We provide order-of-magnitude estimates for the two most accessible predictions.

Frame-transformation hysteresis. Consider a qubit coupled dispersively to a readout resonator (dispersive shift $\chi/2\pi \sim 1$ MHz). Prepare the state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, drive the which-path parameter through a cycle $\lambda : 0 \to \lambda_{\max} \to 0$ by pulsing the readout, then measure visibility. In standard QM with a closed erasure cycle, $\mathcal{V}_{\text{revival}} = \mathcal{V}_{\text{initial}}$. In HCR, the action cost of the cycle is

$$\Delta\mathcal{S} = \frac{1}{4D} \oint (\dot{\lambda} - F^\lambda)^2 G_{\lambda\lambda}\, d\sigma > 0, \tag{25}$$

The action cost $\Delta\mathcal{S} > 0$ suppresses the probability of path-reversed (revival) trajectories relative to forward (erasure) trajectories by a factor $\sim e^{-\Delta\mathcal{S}}$ in the weak-noise limit, which translates to a visibility deficit $\delta\mathcal{V} \sim \Delta\mathcal{S}$ for small $\Delta\mathcal{S}$. Evaluating the action cost for a symmetric cycle $\lambda : 0 \to \lambda_{\max} \to 0$ gives $\Delta\mathcal{S} \sim \tau_{\text{frame}}^2\, G_{\lambda\lambda}(\bar{\lambda})/D$, where $\bar{\lambda}$ is the characteristic distinguishability traversed and D is the effective diffusion coefficient on Σ.

A dimensional estimate, connecting $D \sim 1/T_2$ and using the Ricci scalar $\mathcal{K}_\Sigma = 8$ of the qubit coherence space (CP^1 with Fubini–Study metric radius $r = 1/2$, giving $\mathcal{K} = 2/r^2 = 8$ [19]), yields a visibility deficit $\delta\mathcal{V} \sim (\tau_{\text{frame}}/T_2)^2 \cdot \mathcal{K}_\Sigma$, where $\tau_{\text{frame}} \sim 1/\chi \approx 1$ μs and $T_2 \approx 70$ μs. This gives $\delta\mathcal{V} \sim 2 \times 10^{-3}$. A rigorous derivation connecting $\Delta\mathcal{S}$ to the visibility deficit via a specific noise model is deferred to Paper 2; the estimate here serves as an order-of-magnitude guide to experimental sensitivity requirements.

We stress that this hysteresis does not require unitarity violation. The full system–environment state evolves unitarily throughout; the visibility deficit $\delta\mathcal{V}$ is a property of the *reduced* dynamics on Σ—the action cost $\Delta\mathcal{S} > 0$ (Eq. 25) accumulated along the erasure–revival cycle under the effective stochastic drift F^λ. In standard QM the path through Σ carries no metric weight, so $\mathcal{V}_{\text{revival}} = \mathcal{V}_{\text{initial}}$. In Coherence Relativity the Fubini–Study metric renders forward and reverse paths geometrically inequivalent, producing the predicted deficit. The distinction is analogous to reversible vs. irreversible thermodynamic paths: global energy conservation holds in both, but accessible work differs.

Transformation timescales. The geodesic distance in Σ between the coherent frame ($\lambda = 0$) and the pointer frame ($\lambda = 1$) sets a minimum transformation duration:

$$\tau_{\text{frame}} = \frac{1}{v_{\max}} \int_0^1 \sqrt{G_{\lambda\lambda}(\lambda)}\, d\lambda, \tag{26}$$

where $v_{\max}$ is bounded by the measurement coupling strength. For a transmon qubit with $\chi/2\pi = 1$ MHz coupled to a single readout resonator ($N = 1$), the metric takes the form $G_{\lambda\lambda} = 1/[2(1-\lambda^2)]$ under the EGY parameterization, and the geodesic integral evaluates analytically to $\int_0^1 \sqrt{G_{\lambda\lambda}}\, d\lambda = \pi/(2\sqrt{2}) \approx 1.1$, giving $\tau_{\text{frame}} \gtrsim 1.1$ μs. For multi-mode environments ($N = 10$, Fig. 1), the integral over the experimentally accessible range evaluates to ≈ 1.5. Both timescales are resolvable via weak-measurement trajectories [53]: monitoring the qubit with a weak probe during the frame transformation should reveal a finite-duration transition rather than an instantaneous jump, with the transition profile reflecting the monotone $G_{\lambda\lambda}(\lambda)$ from Fig. 1. Since the physical decoherence rate satisfies $ds/dt = \sqrt{G_{\lambda\lambda}}\, d\lambda/dt$, the metric directly shapes the temporal trajectory: rapid state evolution near the coherent endpoint (where $G_{\lambda\lambda} \approx 1/2$ is small) and progressively slower passage as classical records are established (where $G_{\lambda\lambda}$ grows large).

Remark 9 (Universality of timescales)**.** *The timescales above are* not *universal predictions of the framework: they depend on the coupling strength $v_{\max}$ that sets the maximum rate of motion through Σ. What* is *universal is the geometric constraint* (26)*—a minimum geodesic distance must be traversed—and the characteristic shape of $G_{\lambda\lambda}(\lambda)$. Different experimental platforms (femtosecond photonic vs. microsecond superconducting) will exhibit different absolute timescales because their couplings differ; the testable content is the* ratio *of transformation duration to coupling strength and the path-dependent structure of the metric.*

G. Relationship to Existing Frameworks

HCR is closest in spirit to relational quantum mechanics [10] but differs in three respects: (i) Σ has concrete geometric structure; (ii) the Born rule is derived, not assumed; (iii) the tensor field T_{AB} provides dynamics absent in RQM.

Compared to QBism [11] and Healey's pragmatist quantum theory [55], HCR is realist: frame-invariant content (Born measures, inner products, spectra) constitutes observer-independent physical fact. Healey's approach shares HCR's emphasis on relational structure but does not derive the Born rule or provide a geometric arena.

It is also useful to distinguish coherence relativity from Benoist-style contextual realism in contemporary philosophy [56]. There the core claim is metaphysical and metaphilosophical: reality is neither a mind-dependent construction nor an acontextual "in-itself," but is articulable only within specific language-games, practices, and categorial settings. Coherence relativity shares the rejection of naive acontextual realism, but the sense of context is different. Here context is physical rather than primarily linguistic or normative: boundary conditions, interaction networks, information-bearing channels, and the record structure encoded by a coherence frame. The project is therefore not a Wittgensteinian therapeutic redescription of scientific discourse, but a dynamical and geometric proposal about the physical structure from which effective laws arise.

A related comparison point is the family of contextual realist interpretations of quantum mechanics associated with authors working near relational and neo-Copenhagen approaches [57]. Those accounts typically leave the standard quantum formalism intact and argue that measurement outcomes and state-ascriptions should be understood as context-dependent while preserving realism about experimentally accessible structures. Coherence relativity overlaps with them in rejecting acontextual hidden-value pictures and in treating definiteness as frame- or context-bound. It departs, however, in ambition and scope: rather than interpreting a fixed formalism, it is proposed as an underlying coherence-first framework from which quantum behavior, local field dynamics, and relativistic spacetime structure emerge as regime-specific limits.

Compared to many-worlds [58, 59], HCR posits one reality and many frames, not many worlds and one agent. The logical structure of the Born rule derivation parallels the Deutsch–Wallace approach, but replaces branching ontology with geometric frame-invariance. On this reading, the branching structure ordinarily associated with Everettian multiplicity is relocated from a proliferation of spacetime worlds to the space of admissible decohering continuations on Σ: different quasi-classical histories correspond to different frame-relative sectors and transport paths available within the common coherence manifold. Emergent spacetime $\mathcal{M}$ then records a realized quasi-classical history rather than serving as the fundamental container of all branches. In this sense, HCR preserves continuous Schrödinger evolution of the total state while re-locating branching and record formation from a multiplicity of worlds to the geometry of coherence frames. This is an interpretive consequence of the kinematics developed here; a full dynamical account of how particular spacetime histories are selected on $\mathcal{M} \times \Sigma$ is deferred to Paper 2.

Compared to Bohmian mechanics [60], HCR also assigns explanatory weight to a geometrically defined structure guiding the emergence of classical behavior, but the role of the quasipotential is different. Bohm's quantum potential is defined in configuration space through a polar decomposition of ψ and guides hidden particle trajectories; the HCR quasipotential $V(\lambda)$ is derived from the coordinate-invariant Fubini–Study geometry and guides frame evolution on Σ toward stable pointer sectors. The analogy is therefore structural rather than ontological: HCR retains the guidance-landscape intuition without introducing hidden positions or a second ontology alongside the wavefunction.

The quantum reference frame (QRF) program [61–63] shares HCR's emphasis on frame dependence. In QRF work, entanglement and superposition can change when the reference system changes [62], and several approaches seek perspective-neutral structures [64, 65]. HCR differs in scope. QRFs transform between quantum systems serving as references, whereas coherence frames transform between *descriptive bases* for a fixed system. The coherence space Σ supplies a global geometric arena that is usually absent in QRF treatments, and in HCR the Born rule appears as the unique invariant measure on that arena rather than as an extra assumption. Recent work showing that the sum of entanglement and subsystem coherence is invariant under QRF transformations [54] supports the broader claim that physical reality is carried by frame-invariant quantities, not by any one frame-dependent description. Figure 10 summarizes the relation.

The relational flavor of HCR is especially close to Rovelli's loop quantum gravity program [66]. In that vocabulary, spin networks are discrete relational states

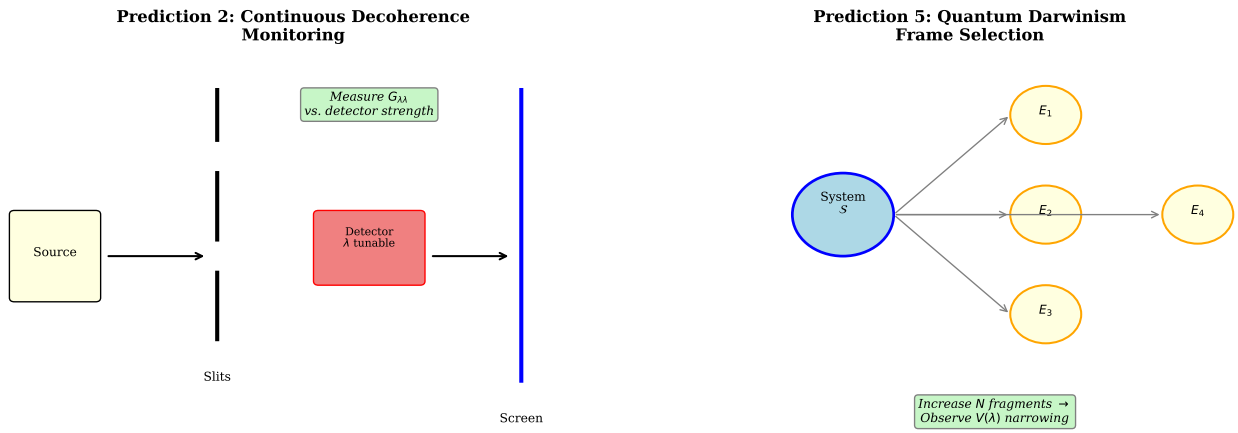

FIG. 9. Experimental protocols for the two most accessible HCR predictions. **(a)** Prediction 2 (frame-transformation hysteresis): a tunable-strength detector between the slits and screen allows direct measurement of $G_{\lambda\lambda}$ as a function of detector coupling χ. A complete erasure cycle $\lambda : 0 \to \lambda_{\max} \to 0$ should show path-dependent visibility recovery $\delta\mathcal{V} \sim (\tau_{\text{frame}}/T_2)^2 \cdot \mathcal{K}_\Sigma$ forbidden in standard QM. **(b)** Prediction 5 (transformation timescales / Quantum Darwinism frame selection): as the number of independently monitoring environment fragments N increases, weak-measurement trajectories probe a sharpening $V(\lambda)$ minimum (Fig. 7) and a slower pointer-frame approach—a longer $\tau_{\text{frame}} \propto \sqrt{N}$ determined by the multi-mode geodesic integral (26).

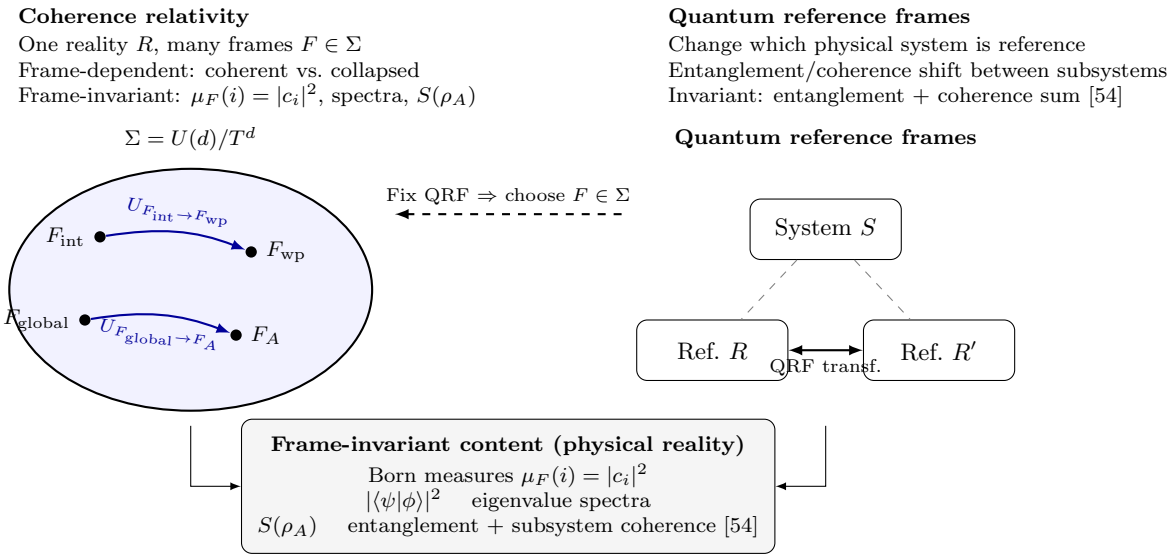

FIG. 10. Coherence relativity vs. quantum reference frames. Left: the coherence manifold Σ with frames F connected by transformations $U_{F\to F'}$. Coherence and collapse are frame-dependent; Born measures and spectra are invariant. Right: QRF transformations change which physical system serves as reference, redistributing entanglement and coherence between subsystems. Dashed arrow: given any QRF choice, one still selects a coherence frame $F \in \Sigma$ on the resulting Hilbert space. Bottom: physical reality consists of quantities invariant under both types of transformation.

of geometry, partial observables are quantities defined through their correlations, and relational time means time read from those correlations rather than from an external clock. In LQG, geometry is not a fixed background but part of the quantum state itself, built from relations among degrees of freedom. HCR adopts the same style of reasoning at the coherence level: coherence frames are not imposed on top of a pre-given quantum state but are defined together with that state through the pointer structure. Rovelli's account of time in terms of correlations among observables [67] likewise parallels the coherence-relativity discussion of past, present, and future in Sec. VII. A fuller unification of LQG-style relational geometry with the joint manifold $\mathcal{M}\times\Sigma$ is reserved for Paper 2.

H. Geometric Emergence Programs

Coherence relativity belongs to a broader contemporary movement in which geometric objects are more fundamental than the physical principles they encode. The Amplituhedron program of Arkani-Hamed and Trnka [68] encodes scattering amplitudes in N=4 Super Yang-Mills theory as the volume of a purely geometric object in Grassmannian space—with no Feynman diagrams, no spacetime, and without assuming locality or unitarity. These emerge from the geometry. The structural parallel to HCR is direct: just as the Born rule emerges here as the unique invariant measure on Σ rather than being postulated, unitarity emerges from the Amplituhedron rather than being assumed. Both programs suggest that the deepest level of physical description is geometric, and that the principles physicists have long taken as foundational are in fact consequences of geometric structure. The connection between the coherence manifold Σ and the positive Grassmannian underlying the Amplituhedron is an open question warranting future investigation.

I. Temporal Structure and the SR–QM Convergence

The coherence parameter λ also suggests a natural temporal reading. At $\lambda = 0$ the system is fully coherent: no stable classical record has formed, and multiple outcomes remain open possibilities. As $\lambda \to 1$, the system approaches a pointer frame: records become stable, outcomes become definite, and the metric $G_{\lambda\lambda}$ diverges. We interpret this divergence as the rigidity of the past. Once a classical record is fully established, reversing the trajectory through Σ becomes geometrically costly.

To keep these temporal claims distinct, three structures should be separated. $\mathcal{M}$ *alone*: ordinary spacetime ordering of events, with the familiar SR relativity of simultaneity for spacelike separation. Σ *alone*: not a second clock, but the ordering of decoherence progress—how far record formation has advanced along a trajectory in coherence space. $\mathcal{M} \times \Sigma$: the full HCR description, in which an interaction has both a spacetime location and a coherence-frame status telling us whether its associated records are still open, still forming, or already settled. The key point is that coordinate-time order and fact-closure on Σ need not coincide.

This suggests a simple three-part picture inside HCR. Coherence ($\lambda \approx 0$) corresponds to the open future, where no observer-independent fact about a measurement outcome has yet been fixed. The intermediate region $0 < \lambda < 1$ corresponds to mixed, partly decohered frames: the present, where records are in the process of being formed. Decoherence ($\lambda \to 1$) corresponds to the fixed past, where stable pointer records satisfy Proposition 1. Because $G_{\lambda\lambda} \to \infty$ as $\lambda \to 1$ (Sec. II D), the geodesic bound (26) implies that this transition cannot be literally instantaneous.

Operationally, the temporal labels line up with frame type as follows:

Frame	λ	Record status	HCR-time
Coherent	≈ 0	no stable records	future
Mixed/ intermediate	$0 < \lambda < 1$	records forming	present
Classical/ pointer	$\to 1$	stable records	past

The Born measure $\mu_F(i) = |c_i|^2$ exists throughout this process. It is defined even in fully coherent frames, where $\mathcal{D}_F \approx \emptyset$. For that reason the HCR-future is better described as pre-record than pre-Born. The weights are already there. What is missing is the stable record structure that turns those weights into definite facts about particular outcomes. Put differently: decoherence does not create the Born weights; it stabilizes the records that instantiate them.

Retrocausal paradoxes dissolved.—This viewpoint also removes the force of retrocausal paradoxes. Delayed-choice and some weak-measurement experiments seem to show that a later intervention changes an earlier fact. In HCR, the key distinction is between coordinate-time order and HCR-temporal order. An event is HCR-past only when it has become a Born-compliant, observer-independent fact across the relevant transport loops. Roughly speaking, the issue is whether the candidate fact survives a closed cycle of frame changes and comes back unchanged. An event can therefore be earlier in coordinate time and still remain in the HCR-present for some frames. The later intervention does not change the past; it helps complete the transition through Σ by which the fact becomes globally fixed.

Convergence of SR and QM temporal relativity.—This is the point of contact with special relativity. SR says that simultaneity is frame-dependent. HCR says that the status of an event as future, present, or past is frame-dependent at the quantum level. In both cases, temporal relations are not absolute at the relevant level of description. Just as a Lorentz boost can change which distant events count as simultaneous without changing the underlying events, a coherence-frame change can alter whether a record counts as still forming or already settled without changing the underlying process. On the spacetime side, the dependence is governed by the Lorentz group $O(1,3)$. On the coherence side, it is governed by the group G of Born-preserving coherence-frame transformations. The joint manifold $\mathcal{M} \times \Sigma$ is therefore the natural place to state the shared principle. Full geometric unification, including dynamics on $\mathcal{M} \times \Sigma$ and holographic connections, is deferred to Paper 2.

Minimal ontology. To summarize the interpretive commitments: HCR posits *one reality* $\mathcal{R}$, *many coherence*

frames $F \in \Sigma$, and *frame-invariant quantities as physical facts.* Collapse, coherence, and classicality are frame-dependent descriptions; Born measures, inner products, spectra, and entanglement entropy are frame-invariant and constitute observer-independent reality.

VII. CONCLUSION

We have introduced coherence relativity as the coherence-frame sector of Holographic Coherence Relativity, a geometric framework in which:

1. Quantum coherence is frame-relative, like simultaneity in special relativity.

2. The Born rule $\mu_F(i) = |c_i|^2$ is the unique frame-invariant probability measure on the coherence manifold Σ.

3. Wave function collapse is a frame transformation, not a dynamical process.

4. The information geometry of Σ, encoded in the Fubini–Study metric and the tensor field T_{AB}, generates a quasipotential landscape explaining einselection and classical emergence.

5. Temporal ordering—past, present, and future—is coherence-frame-relative: decoherence corresponds to the fixed past ($\lambda \to 1$, $G_{\lambda\lambda} \to \infty$), coherence to the open future ($\lambda \approx 0$), and the frame-transition zone to the present. HCR and SR thereby instantiate the same temporal-relativity principle at different strata of physical description (Sec. VII).

The framework unifies insights from Gleason, Zurek, and operational reconstruction programs while identifying eight experimentally testable signatures. The most accessible—frame-transformation hysteresis and measurable transformation timescales—are within reach of current superconducting qubit and trapped ion experiments.

Paper 2 will develop the full tensor field T_{AB} on $\mathcal{M} \times \Sigma$, including the explicit derivation of $T_{M\Sigma}$, the $\delta\lambda$ formalism separating frame-lag from decoherence progress, the coordinate-invariant Markov transition criterion, the mixed-states extension of the Born rule via purification, equations of motion for frame dynamics, and holographic connections between coherence-space geometry and emergent spacetime. Paper 2 further develops a dynamical derivation of the Born rule from the self-consistency fixed-point condition on $\mathcal{M} \times \Sigma$ (the SCF→COV→noncontextuality→Gleason/Busch chain), recovering the present axiomatic result (Theorem 2) as a consequence in the SCF-saturated sector; the holographic connections developed in Paper 2 are completed in the Paper 3 framework, where $\mathcal{M}$ is derived from $\mathcal{H}$.

DECLARATIONS

Use of Artificial Intelligence Tools

This manuscript was developed with substantive assistance from multiple large language model (LLM) systems operating as an iterative, coordinated research ensemble. The AI systems involved were Claude (Anthropic), ChatGPT (OpenAI), Perplexity AI, and Warp AI, each contributing according to their respective functional strengths across different phases of the work. To facilitate structured communication and task coordination among these AI systems, the author developed a purpose-built inter-AI communication protocol (the Society of Mind conversation log, "conv-log") specifically for the execution of this paper series; this protocol enabled persistent cross-session coordination, iterative review handoffs, and traceable research state across the multi-agent ensemble.

A formal quality-control framework governed all AI participation throughout the research process. Explicit operational ground rules were established and enforced across all AI systems, requiring: (i) strict calibration of confidence levels to the actual state of evidence, with mandatory distinction between derived results, working assumptions, and speculative claims; (ii) proactive disclosure of uncertainty, data limitations, and calculation issues at the time of occurrence rather than post hoc; and (iii) aggressive multi-AI cross-validation at every phase, in which results, derivations, and arguments produced by one AI system were independently reviewed and challenged by others before being accepted. This architecture was designed to counteract the known failure modes of individual AI systems—overconfidence, hallucination, and uncritical propagation of errors—and to ensure that outputs entering the manuscript had survived structured adversarial scrutiny within the ensemble.

Collectively, these tools and the coordination infrastructure provided: iterative review and assessment of theoretical arguments and logical architecture across multiple revision cycles; drafting and refinement of expository passages, including the coherence-frame analogy development, the Born rule derivation narrative, and the SR comparison framework; procedural guidance on mathematical methodology and derivation strategy; literature identification and synthesis relevant to the foundations of quantum mechanics; cross-checking and consultation on mathematical steps; and generation of Python code for the numerical calculations underlying the figures. The collaborative, multi-agent nature of this process, and the novel coordination methodology developed to support it, were integral to the development and refinement of the manuscript rather than incidental to it.

xAI's Grok contributed targeted mathematical review in the March 2026 revision, identifying the EGY parameterization inconsistency and correcting the metric endpoint behaviour (fixes 7–10 of the March 19 revision); this contribution was specific to that revision cycle rather than sustained across the full manuscript development.

All theoretical claims, mathematical derivations, interpretive conclusions, and scientific judgments are solely those of the author. All AI-assisted content was reviewed, verified, and accepted or modified by the author, who takes full responsibility for the accuracy and integrity of the work in its entirety.

Competing Interests

The author declares no competing financial or non-financial interests.

Data Availability Statement

All numerical data supporting the results in this study (Fubini–Study metric calculations, quasipotential landscapes, and figure generation) are available from the author upon reasonable request. The Python scripts used to generate these data are openly available at `https://github.com/bryankaufman/coherence-relativity`. No experimental datasets were generated or analyzed in this theoretical study.

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Author Contributions

B.K. conceived the theoretical framework, performed all calculations, generated all figures, and wrote the manuscript.

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The conceptual foundations of this work rest on contributions from many giants of quantum theory. Von Neumann's mathematical formalization of quantum mechanics [2] established the measurement problem in its modern form. Bohr's insistence on the role of the observer and Einstein's demand for a complete account of physical reality [37] defined the tension this framework addresses. Bell's penetrating analysis of what quantum mechanics actually requires of any physical theory [1] shaped the standard against which interpretations must be evaluated. Everett's relative-state formulation [58] demonstrated that unitary quantum mechanics alone can accommodate the observer, and Gleason's theorem [3] established the mathematical uniqueness of the Born measure. Zurek's decoherence program [4] and Penrose's gravitational collapse proposal [15] represent the two most developed attempts to locate collapse within physics, and both inform the testable predictions made here. The geometric perspective owes much to the information-geometric program developed by Brody and Hughston [20] and to Bengtsson and Życzkowski's synthesis [19]. The author thanks the members of the Society of Mind coordination network—Claude (Anthropic), ChatGPT (OpenAI), Perplexity AI, and Warp AI—for research assistance, computational support, and iterative review throughout manuscript development, and is grateful to the open-source scientific Python community for tools enabling the numerical calculations presented herein. A versioned preprint archive of this manuscript is available at Zenodo via the concept DOI: `https://doi.org/10.5281/zenodo.18675342`.

[1] J. S. Bell, Against 'measurement', in *Sixty-Two Years of Uncertainty: Historical, Philosophical, and Physical Inquiries into the Foundations of Quantum Mechanics*, edited by A. I. Miller (Plenum Press, New York, 1990) pp. 17–31, reprinted in J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics*, 2nd ed. (Cambridge University Press, 2004).

[2] J. von Neumann, *Mathematical Foundations of Quantum Mechanics* (Princeton University Press, Princeton, NJ, 1955) translated by R. T. Beyer from *Mathematische Grundlagen der Quantenmechanik* (Springer, 1932).

[3] A. M. Gleason, Measures on the Closed Subspaces of a Hilbert Space, Indiana Univ. Math. J. **6**, 885 (1957).

[4] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, Rev. Mod. Phys. **75**, 715 (2003).

[5] W. H. Zurek, Probabilities from entanglement, Born's rule from envariance, Phys. Rev. A **71**, 052105 (2005).

[6] D. Deutsch, Quantum theory of probability and decisions, Proc. R. Soc. A **455**, 3129 (1999).

[7] D. Wallace, *The Emergent Multiverse: Quantum Theory According to the Everett Interpretation* (Oxford University Press, Oxford, 2012).

[8] H. D. Zeh, On the interpretation of measurement in quantum theory, Found. Phys. **1**, 69 (1970).

[9] E. Joos, H. D. Zeh, C. Kiefer, D. Giulini, J. Kupsch, and I.-O. Stamatescu, *Decoherence and the Appearance of a Classical World in Quantum Theory*, 2nd ed. (Springer, Berlin, 2003).

[10] C. Rovelli, Relational quantum mechanics, Int. J. Theor. Phys. **35**, 1637 (1996).

[11] C. A. Fuchs, N. D. Mermin, and R. Schack, An introduction to QBism, Am. J. Phys. **82**, 749 (2014).

[12] G. C. Ghirardi, A. Rimini, and T. Weber, Unified dynamics for microscopic and macroscopic systems, Phys. Rev. D **34**, 470 (1986).

[13] P. Pearle, Combining stochastic dynamical state-vector reduction with spontaneous localization, Phys. Rev. A **39**, 2277 (1989).

[14] L. Diósi, A universal master equation for the gravitational violation of quantum mechanics, Phys. Lett. A **120**, 377 (1987).

[15] R. Penrose, On gravity's role in quantum state reduction, Gen. Relativ. Gravit. **28**, 581 (1996).

[16] E. Joos and H. D. Zeh, The emergence of classical properties through interaction with the environment, Zeitschrift für Physik B **59**, 223 (1985).

[17] M. Brune, E. Hagley, J. Dreyer, X. Maître, A. Maali, C. Wunderlich, J. M. Raimond, and S. Haroche, Observing the progressive decoherence of the "meter" in a quantum measurement, Physical Review Letters **77**, 4887 (1996), first quantitative experimental confirmation of decoherence; 26 years after Zeh (1970). Directly verifies decoherence timescale calculations of Joos & Zeh (1985).

[18] S. Haroche, Controlling photons in a box and exploring the quantum to classical boundary, Nobel Lecture, December 8, 2012 (2012), documents the 26-year journey from Zeh's 1970 discovery to experimental confirmation; relevant historical framing for CR Introduction.

[19] I. Bengtsson and K. Życzkowski, *Geometry of Quantum States: An Introduction to Quantum Entanglement*, 2nd ed. (Cambridge University Press, Cambridge, 2017).

[20] D. C. Brody and L. P. Hughston, Geometric quantum mechanics, J. Geom. Phys. **38**, 19 (2001).

[21] T. W. B. Kibble, Geometrization of quantum mechanics, Communications in Mathematical Physics **65**, 189 (1979).

[22] A. Ashtekar and T. A. Schilling, Geometrical formulation of quantum mechanics, in *On Einstein's Path: Essays in Honor of Engelbert Schucking*, edited by A. Harvey (Springer, New York, 1997) pp. 23–65, arXiv:gr-qc/9706069.

[23] D. M. Greenberger and A. Yasin, Simultaneous wave and particle knowledge in a neutron interferometer, Phys. Lett. A **128**, 391 (1988).

[24] B.-G. Englert, Fringe visibility and which-way information: An inequality, Phys. Rev. Lett. **77**, 2154 (1996).

[25] S. L. Braunstein and C. M. Caves, Statistical distance and the geometry of quantum states, Phys. Rev. Lett. **72**, 3439 (1994).

[26] S. Kochen and E. P. Specker, The problem of hidden variables in quantum mechanics, Journal of Mathematics and Mechanics **17**, 59 (1967).

[27] L. Masanes, T. D. Galley, and M. P. Müller, The measurement postulates of quantum mechanics are operationally redundant, Nat. Commun. **10**, 1361 (2019).

[28] B.-H. Chen, D.-W. Chiou, and H.-C. Hsu, Demonstration of Scully–Drühl-type quantum erasers on quantum computers, New Journal of Physics **27**, 074503 (2025).

[29] X. Yang, X. Yu, L. Li, X. Zhao, T. Zheng, C. Zhang, C.-F. Li, and G.-C. Guo, Experimental realization of Wheeler's delayed-choice experiment with dual selections, Science China Physics, Mechanics and Astronomy **68**, 250312 (2025).

[30] E. P. Wigner, Remarks on the mind-body question, in *The Scientist Speculates*, edited by I. J. Good (Heinemann, London, 1961) pp. 284–302, reprinted in E. P. Wigner, *Symmetries and Reflections* (Indiana University Press, 1967).

[31] D. Frauchiger and R. Renner, Quantum theory cannot consistently describe the use of itself, Nature Communications **9**, 3711 (2018).

[32] M. Proietti, A. Pickston, F. Graffitti, P. Barrow, D. Kundys, C. Branciard, M. Ringbauer, and A. Fedrizzi, Experimental test of local observer independence, Science Advances **5**, eaaw9832 (2019).

[33] Č. Brukner, A no-go theorem for observer-independent facts, Entropy **20**, 350 (2018).

[34] J. R. Hance, M. Ji, and H. F. Hofmann, Contextuality, coherences, and quantum cheshire cats, New Journal of Physics **25**, 113028 (2023).

[35] J. R. Hance, J. Ladyman, and J. Rarity, Is the dynamical quantum cheshire cat detectable?, New Journal of Physics **26**, 073038 (2024).

[36] T. Rybotycki, T. Białecki, J. Batle, B. Zglinicki, A. Szereszewski, W. Belzig, and A. Bednorz, Identical, independent quantum weak measurements violate objective realism, arXiv preprint arXiv:2603.22020 10.48550/arXiv.2603.22020 (2026), arXiv:2603.22020.

[37] A. Einstein, B. Podolsky, and N. Rosen, Can quantum-mechanical description of physical reality be considered complete?, Physical Review **47**, 777 (1935).

[38] B. Misra and E. C. G. Sudarshan, The Zeno's paradox in quantum theory, Journal of Mathematical Physics **18**, 756 (1977).

[39] P. Facchi and S. Pascazio, Quantum Zeno subspaces, Physical Review Letters **89**, 080401 (2002).

[40] Y. Li, X. Chen, and M. P. A. Fisher, Quantum zeno effect and the many-body entanglement transition, Physical Review B **98**, 205136 (2018), first demonstration of measurement-induced entanglement phase transition in random quantum circuits.

[41] Y. Bao, S. Choi, and E. Altman, Theory of the phase transition in random unitary circuits with measurements, Physical Review B **101**, 104301 (2020), establishes field-theory description of measurement-induced phase transition (MIPT); maps entanglement transition to replica statistical mechanics.

[42] C. Noel *et al.*, Measurement-induced quantum phases realized in a trapped-ion quantum computer, Nature Physics **18**, 760 (2022), first experimental observation of MIPT in a trapped-ion system; collapse as measurement-induced phase transition with concrete data.

[43] Google Quantum AI, Measurement-induced entanglement and teleportation on a noisy quantum processor, Nature **622**, 481 (2023), experimental MIPT observation on superconducting qubits; directly relevant to CR Prediction 7 timescale regime.

[44] A. G. Kofman and G. Kurizki, Acceleration of quantum decay processes by frequent observations, Nature **405**, 546 (2000).

[45] A. Elga, Self-locating belief and the Sleeping Beauty problem, Analysis **60**, 143 (2000).

[46] P. Winkler, The Sleeping Beauty controversy, The American Mathematical Monthly **124**, 579 (2017).

[47] C. T. Sebens and S. M. Carroll, Self-locating uncertainty and the origin of probability in Everettian quantum mechanics, The British Journal for the Philosophy of Science **69**, 25 (2018).

[48] M. I. Freidlin and A. D. Wentzell, *Random Perturbations of Dynamical Systems*, 3rd ed. (Springer, New York, 2012).

[49] K. Ptaszynski and M. Esposito, Thermodynamics of decoherence, Proceedings of the Royal Society A **479**, 10.1098/rspa.2023.0040 (2023).

[50] W. H. Zurek, Quantum Darwinism, Nat. Phys. **5**, 181 (2009).

[51] C. J. Riedel, W. H. Zurek, and M. Zwolak, The Objective Past of a Quantum Universe, Phys. Rev. A **93**, 032126 (2016).

[52] J. K. Korbicz, P. Horodecki, and R. Horodecki, Objectivity in a Noisy Photonic Environment, Phys. Rev. Lett. **118**, 150503 (2017).

[53] M. Carlesso, A. Bassi, M. Paternostro, and H. Ulbricht, Testing the large-scale limit of quantum mechanics, New J. Phys. **22**, 022001 (2020).

[54] C. Cepollaro, A. Akil, P. Cieśliński, A.-C. de la Hamette, and Č. Brukner, Sum of entanglement and subsystem coherence is invariant under quantum reference frame transformations, Phys. Rev. Lett. **135**, 010201 (2025).

[55] R. Healey, *The Quantum Revolution in Philosophy* (Oxford University Press, Oxford, 2017).

[56] J. Benoist, *Toward a Contextual Realism* (Harvard University Press, Cambridge, MA, 2021).

[57] F.-I. Pris, The real meaning of quantum mechanics, arXiv preprint arXiv:2107.10666 10.48550/arXiv.2107.10666 (2021), arXiv:2107.10666.

[58] H. Everett, Relative State Formulation of Quantum Mechanics, Rev. Mod. Phys. **29**, 454 (1957).

[59] B. S. DeWitt and N. Graham, eds., *The Many-Worlds Interpretation of Quantum Mechanics* (Princeton University Press, Princeton, 1973).

[60] D. Bohm, A suggested interpretation of the quantum theory in terms of "hidden" variables. I, Physical Review **85**, 166 (1952).

[61] S. D. Bartlett, T. Rudolph, and R. W. Spekkens, Reference frames, superselection rules, and quantum information, Reviews of Modern Physics **79**, 555 (2007).

[62] F. Giacomini, E. Castro-Ruiz, and Č. Brukner, Quantum mechanics and the covariance of physical laws in quantum reference frames, Nature Communications **10**, 494 (2019).

[63] A.-C. de la Hamette and T. D. Galley, Quantum reference frames for general symmetry groups, Quantum **4**, 367 (2020).

[64] A. Vanrietvelde, P. A. Höhn, F. Giacomini, and E. Castro-Ruiz, A change of perspective: Switching quantum reference frames via a perspective-neutral framework, Quantum **4**, 225 (2020).

[65] L. Loveridge, T. Miyadera, and P. Busch, Symmetry, reference frames, and relational quantities in quantum mechanics, Foundations of Physics **48**, 135 (2018).

[66] C. Rovelli, *Quantum Gravity* (Cambridge University Press, 2004).

[67] C. Rovelli, Partial observables, Physical Review D **65**, 124013 (2002).

[68] N. Arkani-Hamed and J. Trnka, The amplituhedron, Journal of High Energy Physics **2014**, 30 (2014), arXiv:1312.2007.